Lab Meeting

Objectives

Assess how regulatory sequence and local context shape predicted chromatin accessibility and transcription.

- **Validate predictions**

Compare predicted accessibility with experimental ATAC measurements.

- **Test sequence dependence**

Use shuffling and mutation to probe motif, flanking-sequence and peak effects.

- **Compare regulatory classes**

Assess Stable and Dynamic responses in cardiomyocytes and fibroblasts.

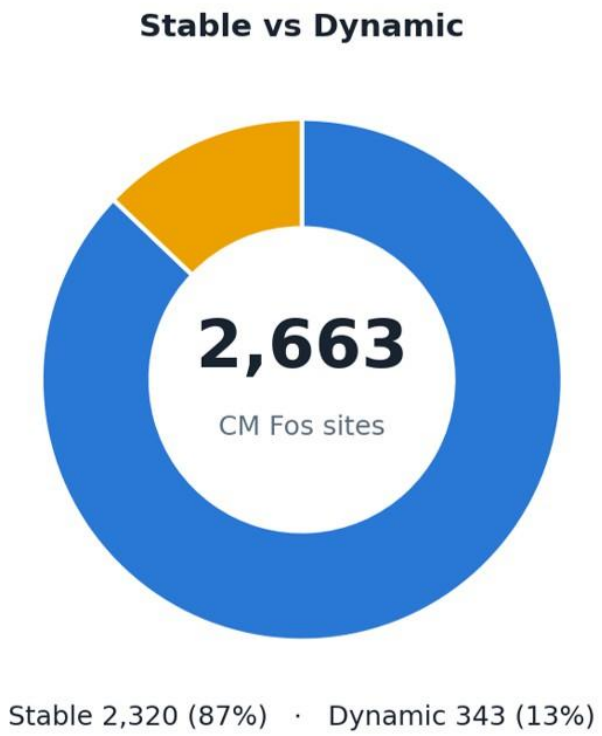
- **Examine chromatin RNA coupling**

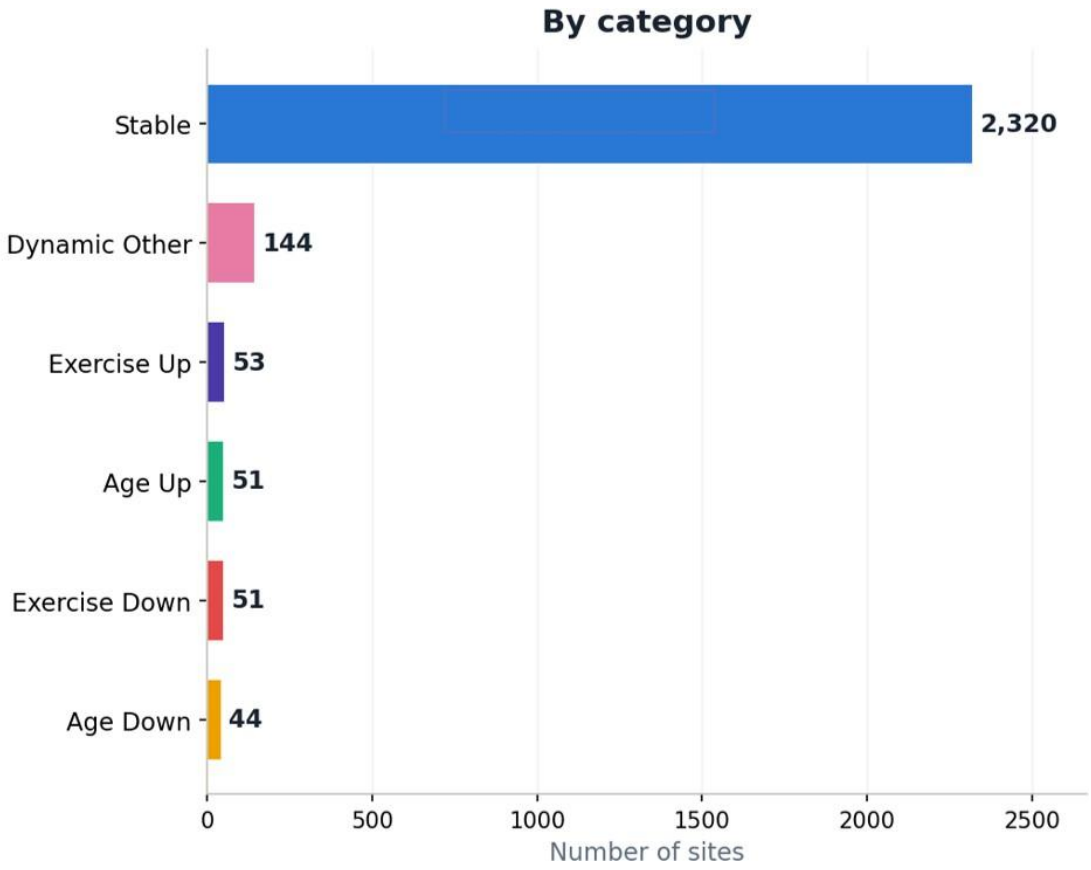
Test whether predicted accessibility changes track regional RNA changes.

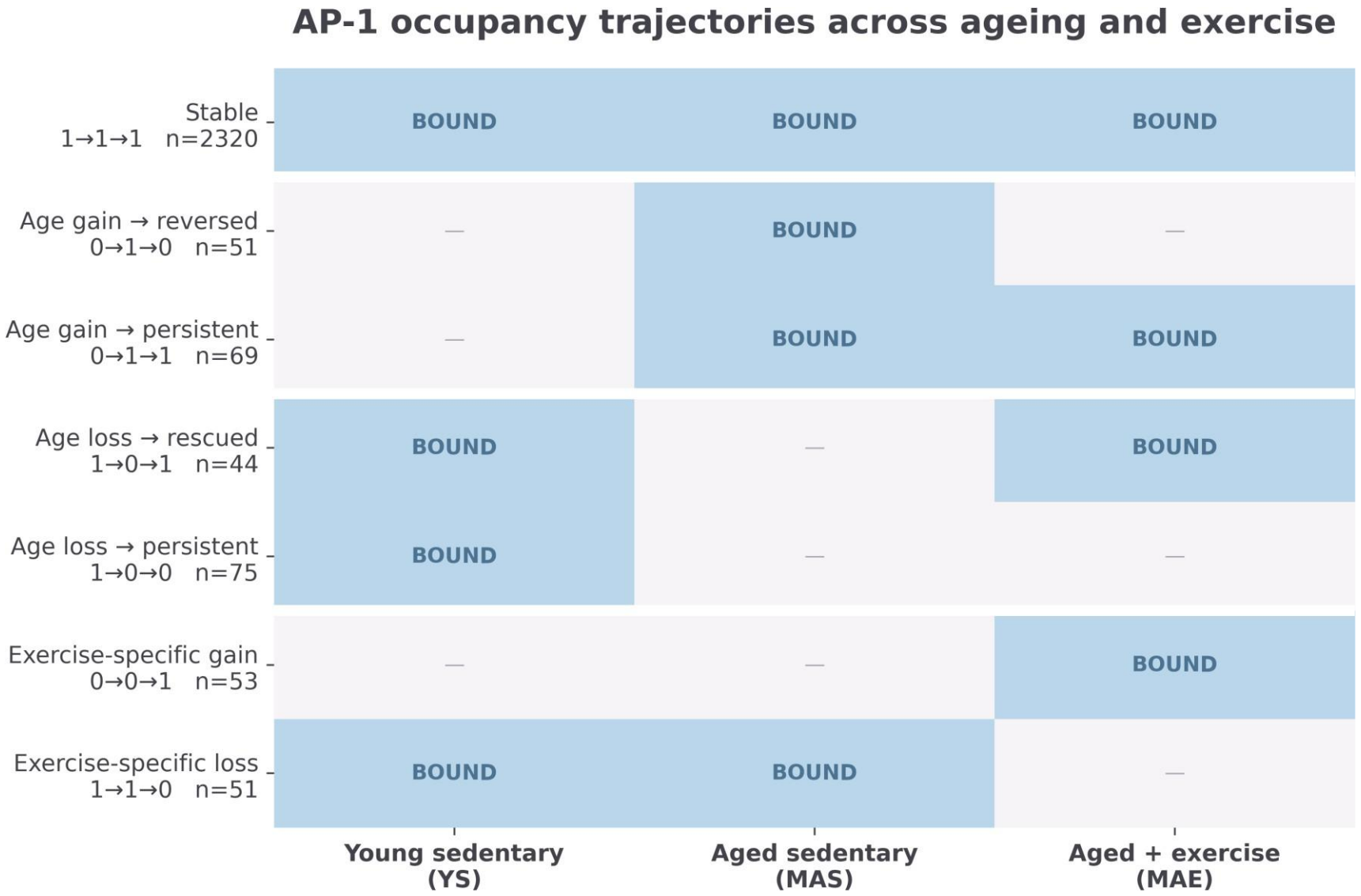
- **Test transfer across genomic contexts**

Insert donor sequences into recipient regions to assess predicted portability.

Dataset Properties when Cardiomyocyte, Fos sites filtered





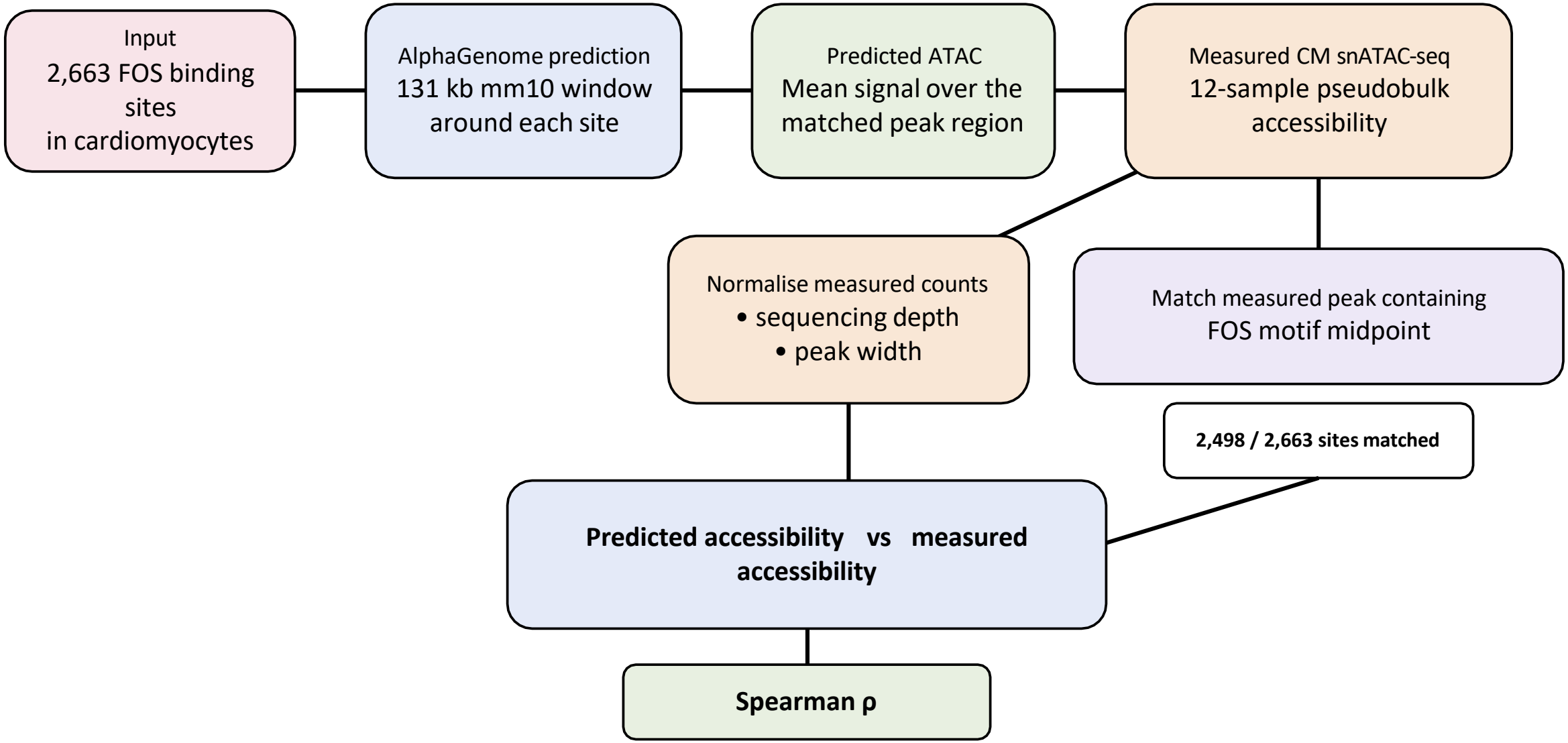


YS vs MAS (Ageing remodelling)

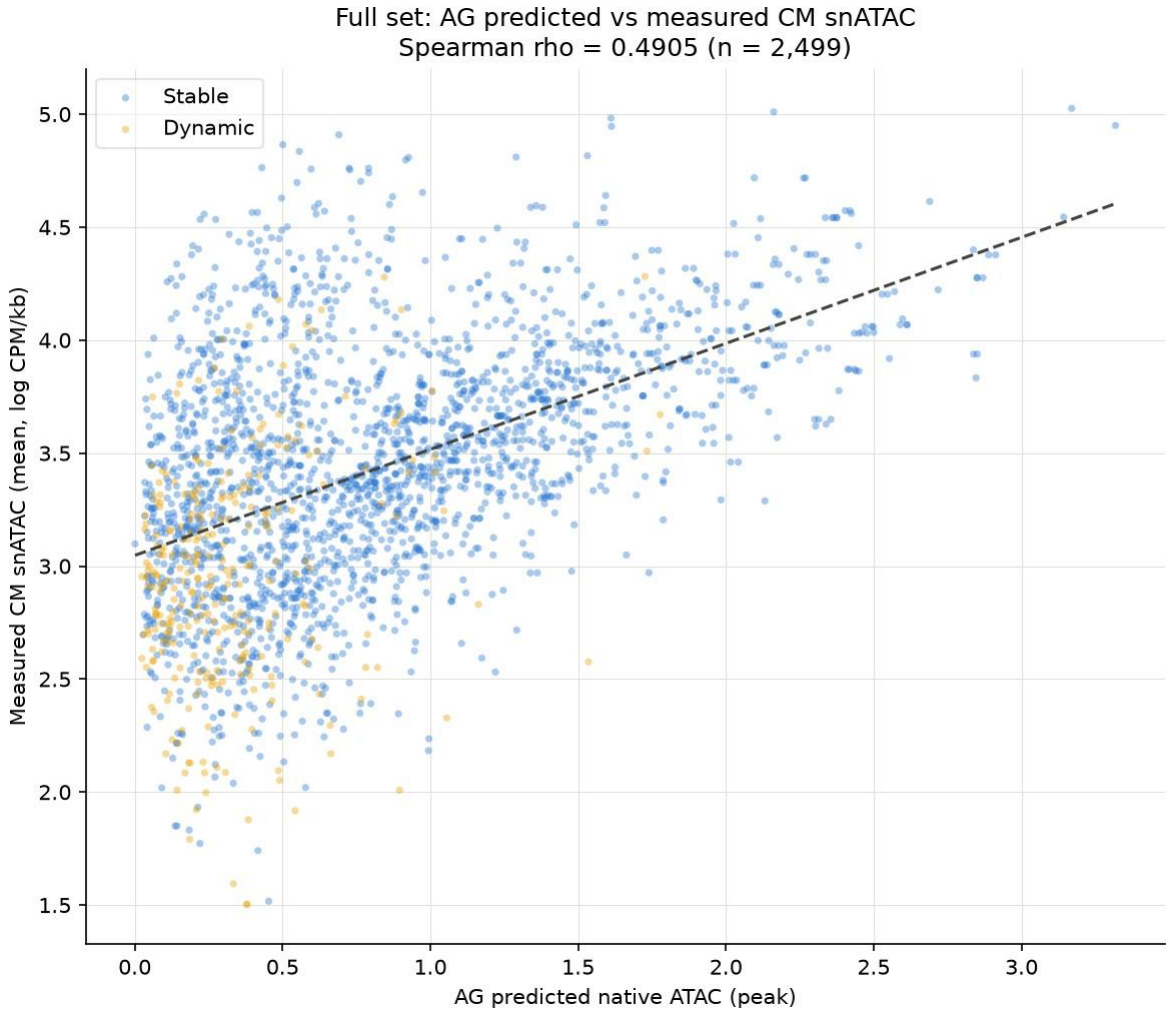
MAS vs MAE (Exercise remodelling)

VALIDATION OF OUTPUT TRACKS OF ALPHAGENOME

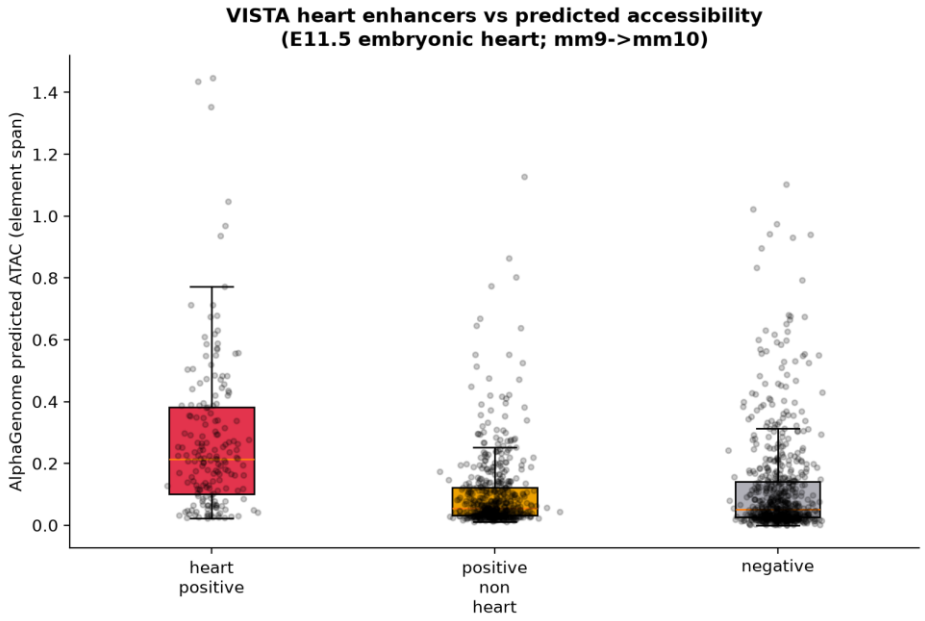
Methodology

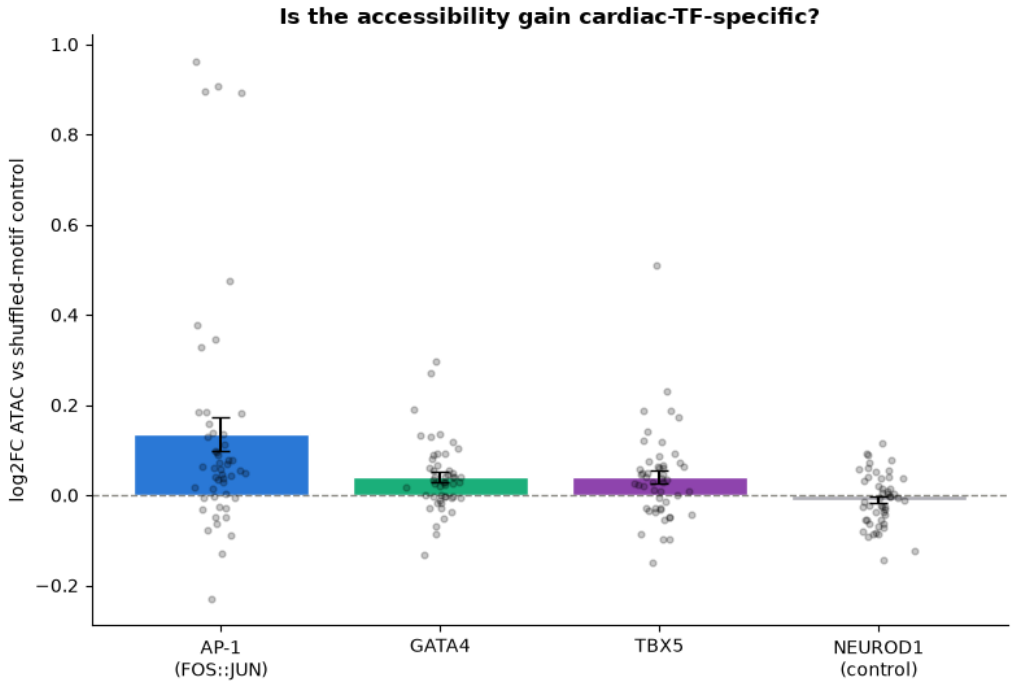


AlphaGenome predictions match measured accessibility

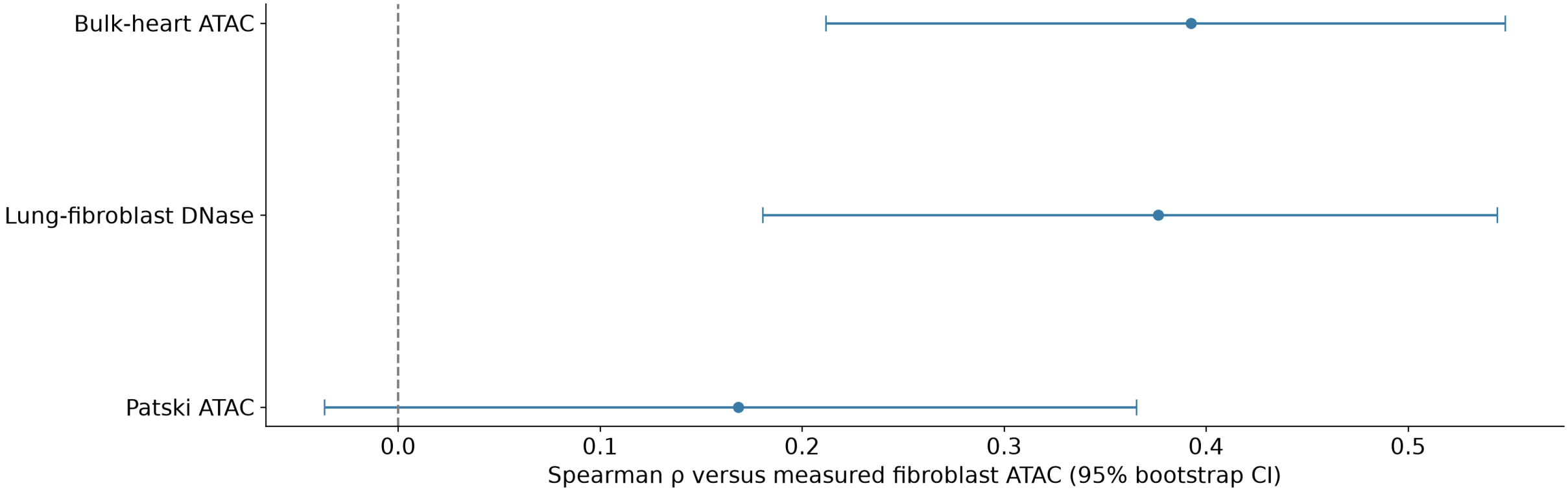


Validation: Heart-positive VISTA elements have higher predicted ATAC





Patski- mouse embryonic kidney fibroblast



Fibroblast validation: 100 sites (76 Stable, 24 Dynamic). Tissue and assay proxies.

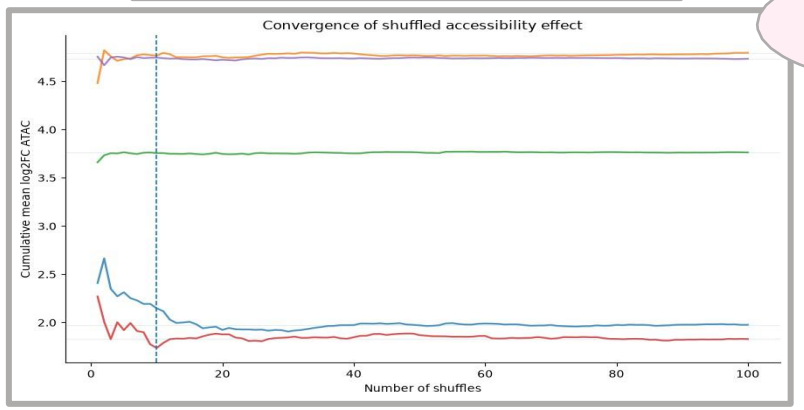
SHUFFLING PERTURBATIONS

METHODOLOGY: Shuffle arm

Input Dataset: Fos binding sites in CM, fibroblasts. Classified as Stable and Dynamic (Age Up/Down, Exercise Up/down, other)

- Motif
- Peak
- 170bp

10 Shuffles
median error vs the 100-shuffle estimate = 2.1%



Motif Shuffle

170bp Shuffle

Peak Shuffle

AlphaGenome Predictions

Accessibility Track

Motif readout
170 bp readout
Peak readout

RNA Track

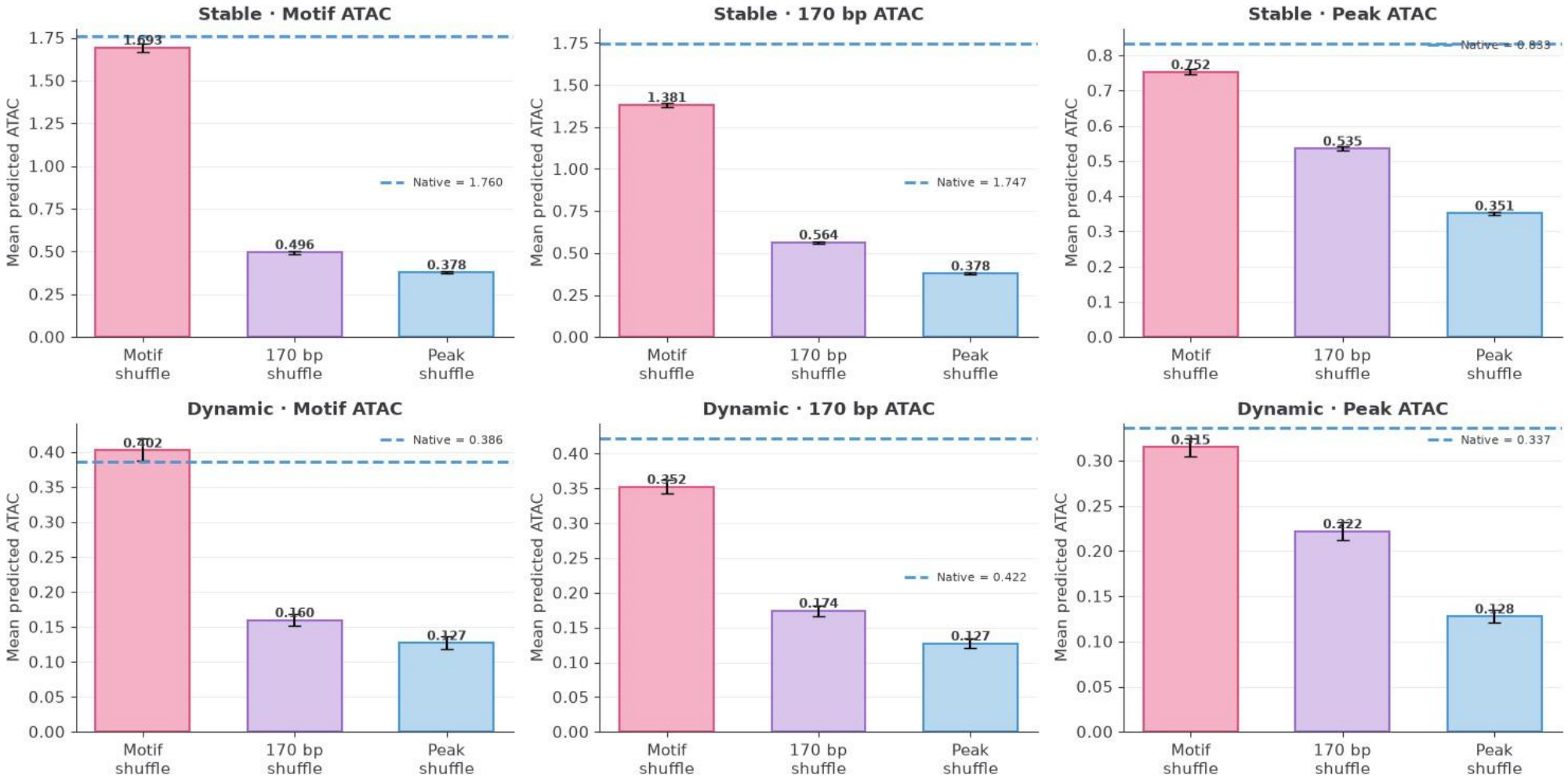
Nearest gene
Mean ±50 kb
Max ±50 kb

Log2FC = Log2 shuffled/native
Aggregate mean, median, SD across 10 shuffles

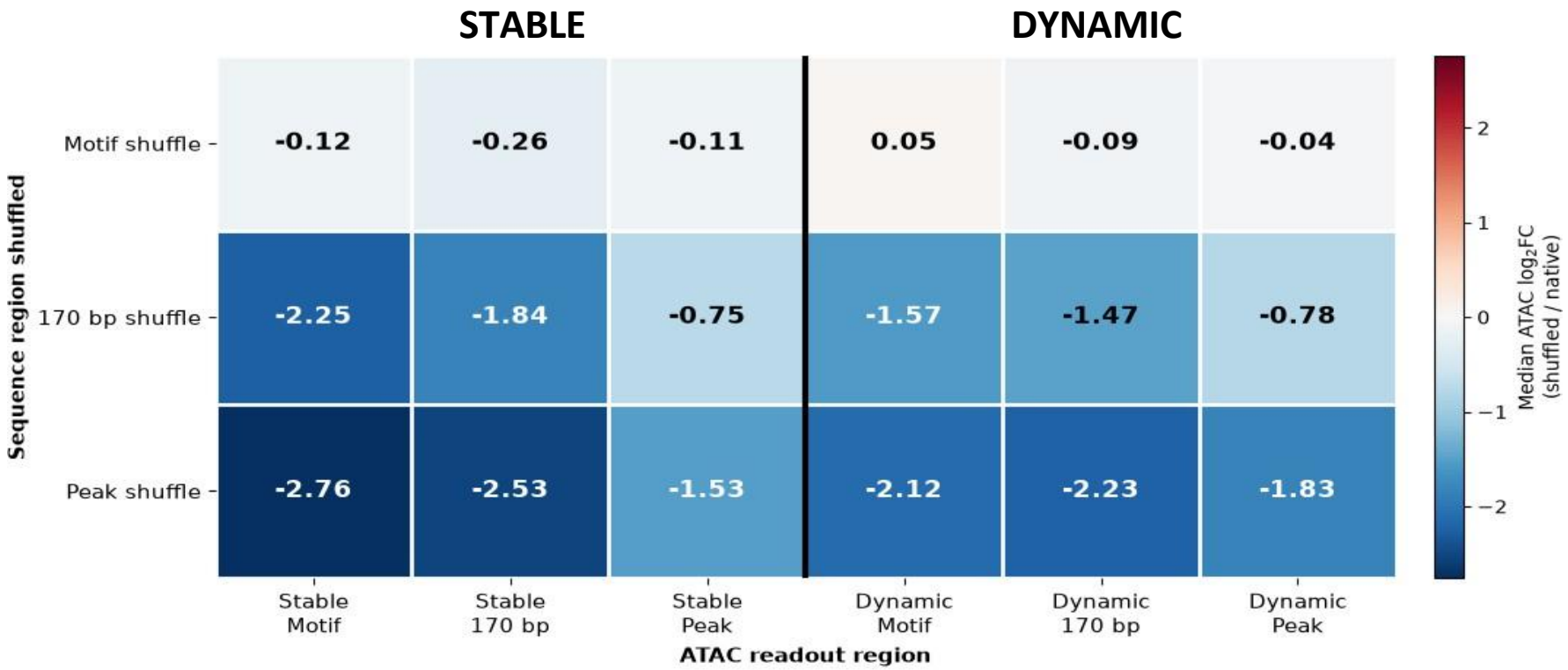
Downstream: Six Class comparisons

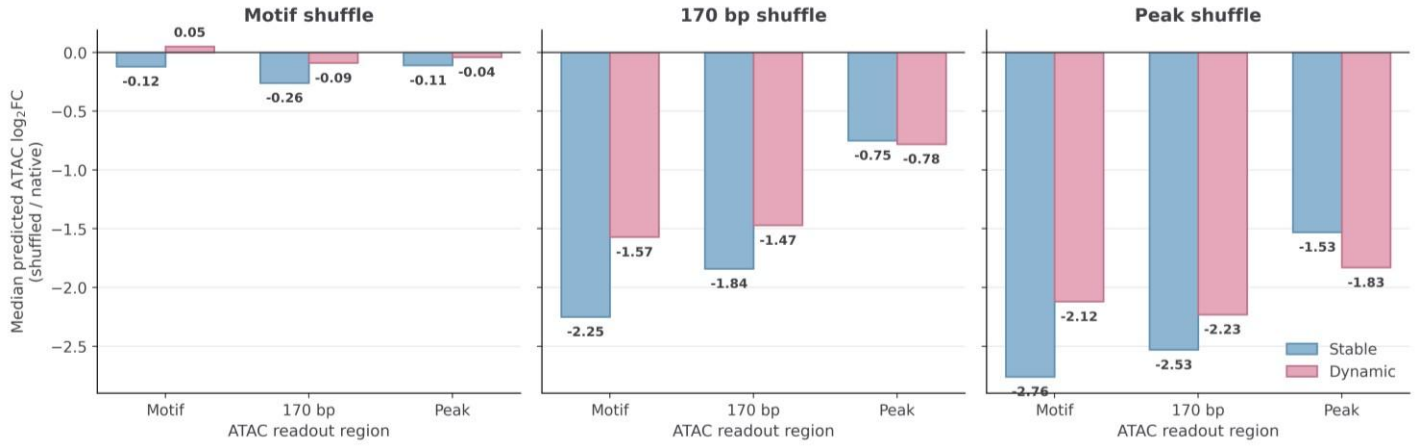
sequence perturbation → ATAC consequence → RNA consequence.

QC : Shuffled < Native signal



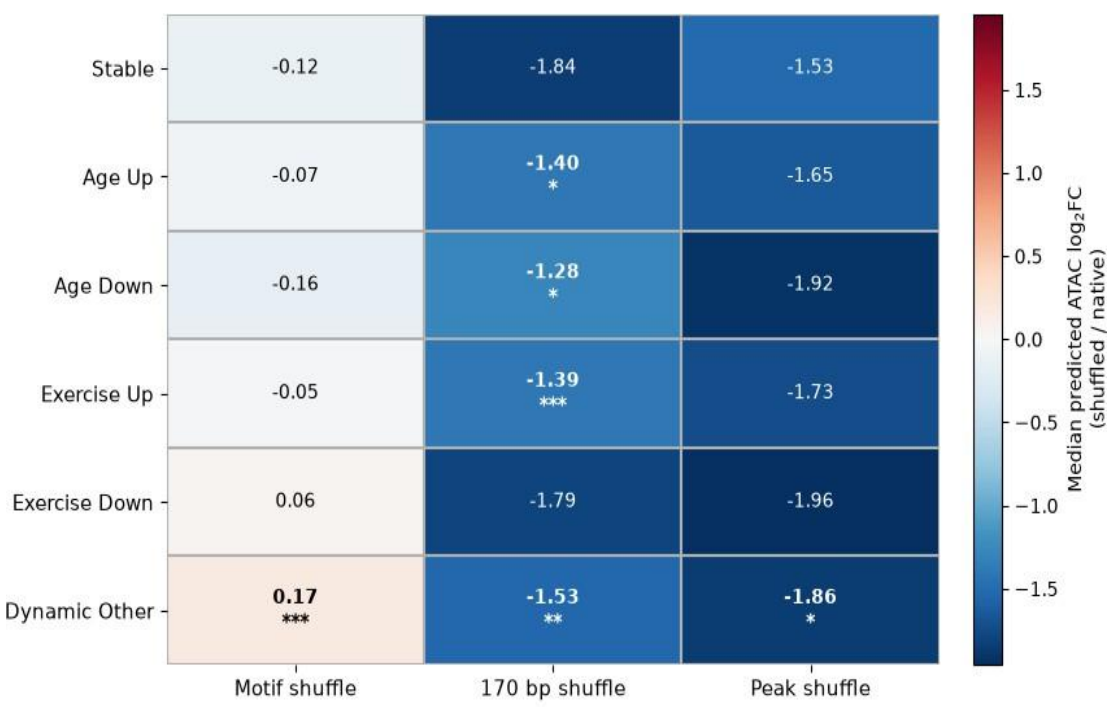
ATAC Readout depends on scale of readout and perturbation



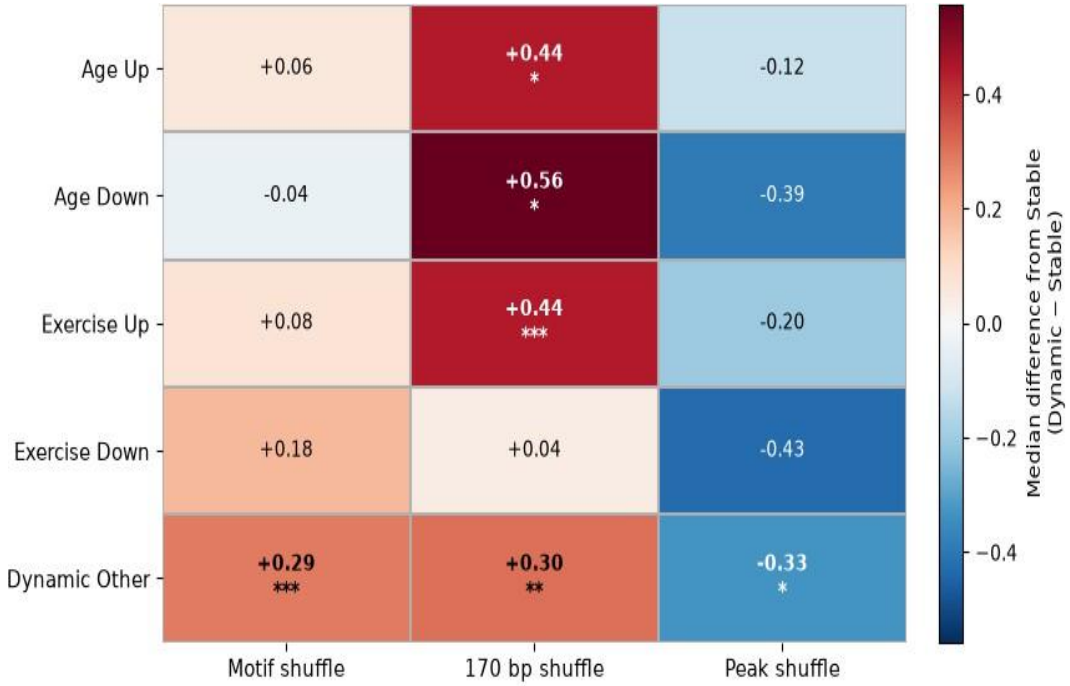


More negative values indicate greater predicted accessibility loss after sequence shuffling.

Motif shuffle → motif ATAC readout
170-bp shuffle → 170-bp ATAC readout
Peak shuffle → peak ATAC readout

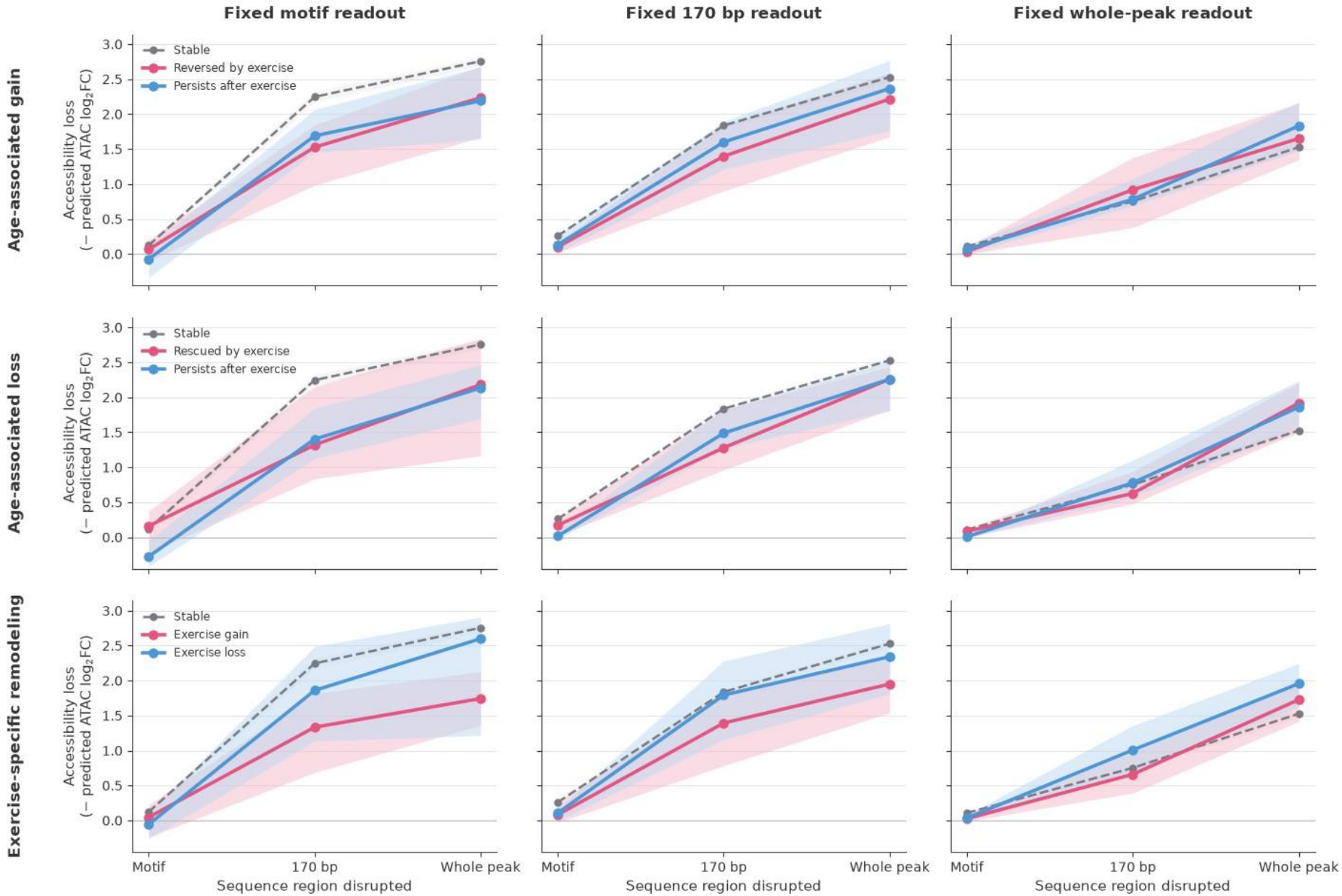


* FDR < 0.05 ** FDR < 0.01 *** FDR < 0.001
Stars indicate comparison with Stable sites; BH correction across 15 tests.

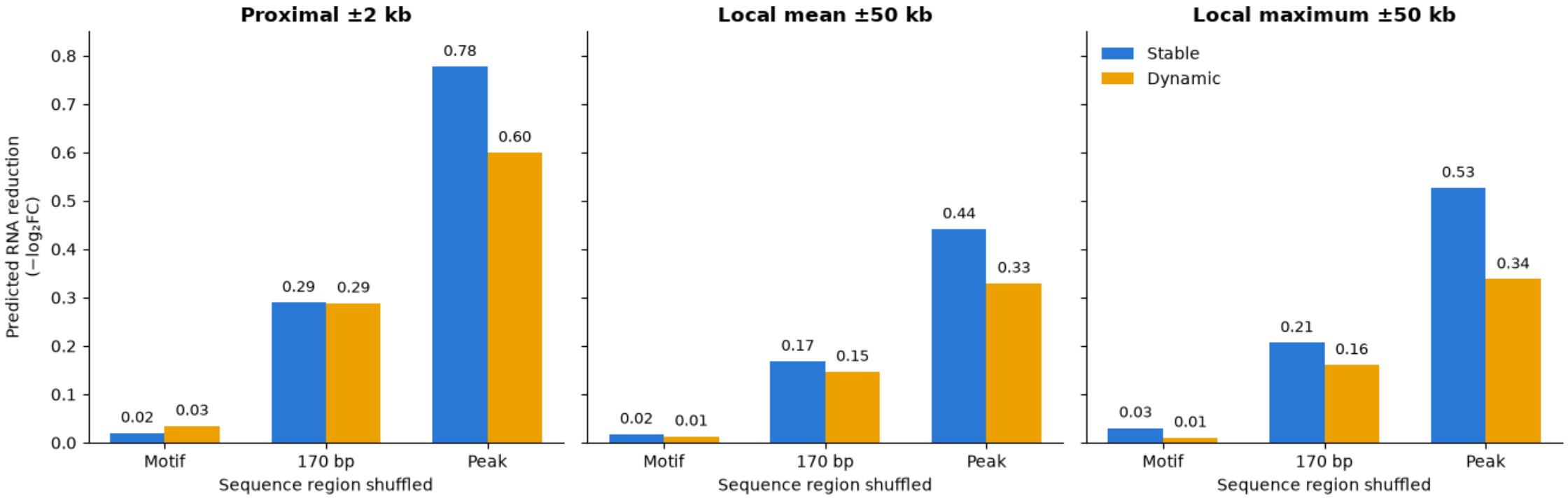


Positive = Dynamic class is less negatively affected than Stable
Negative = Dynamic class is more negatively affected than Stable

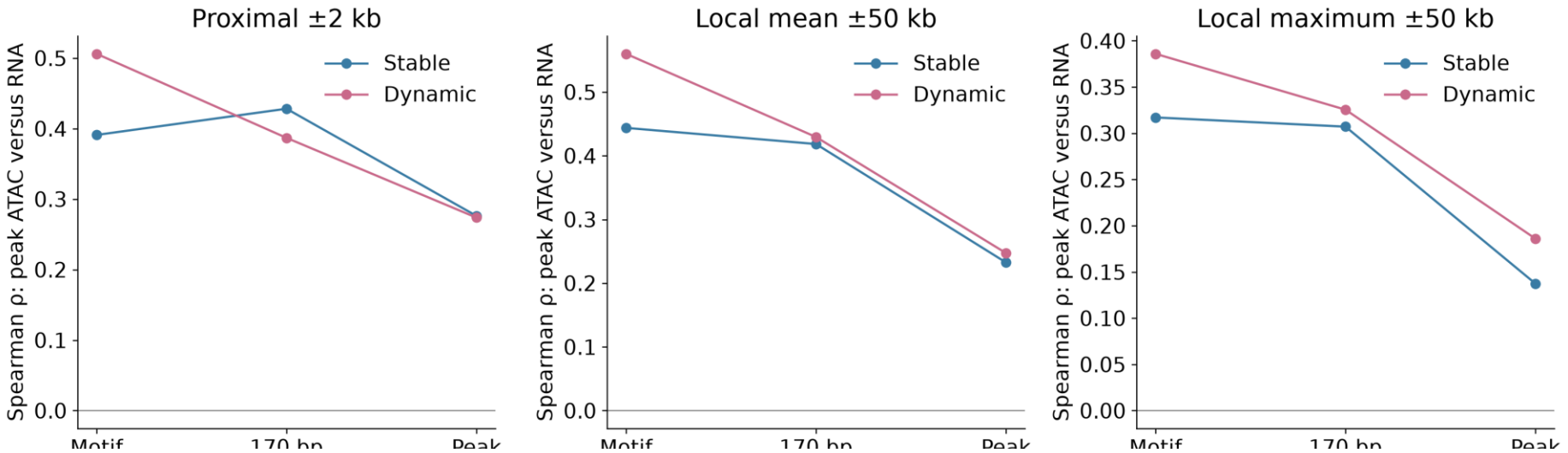
AP-1 accessibility is increasingly dependent on broader surrounding sequence

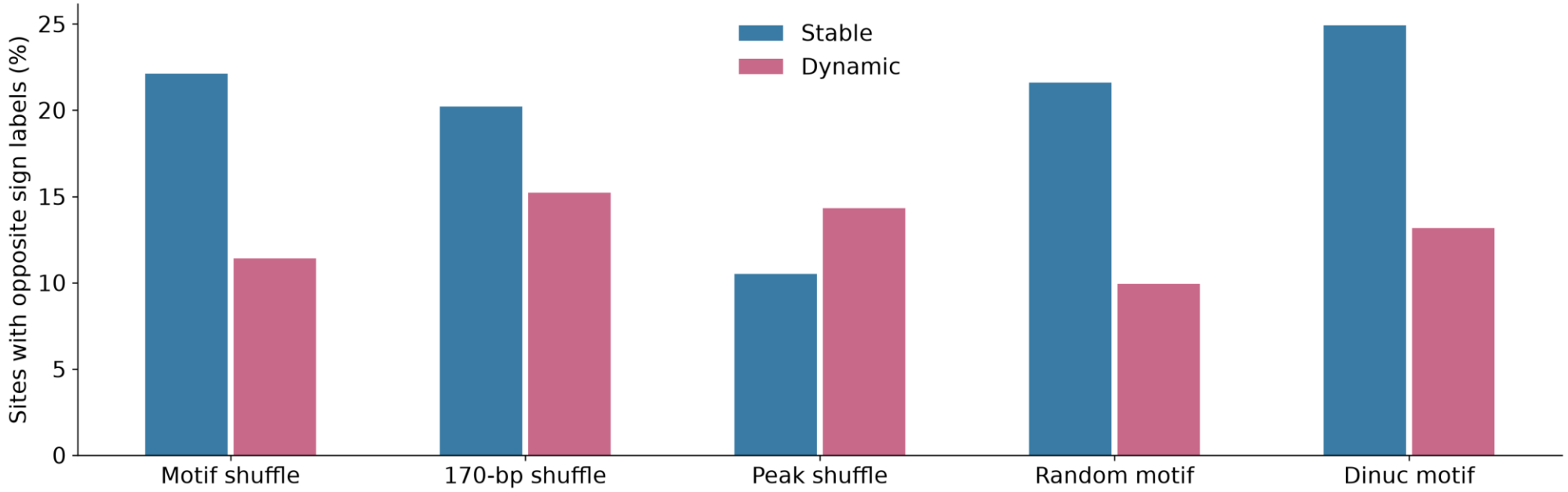


RNA Track- Stable and Dynamic

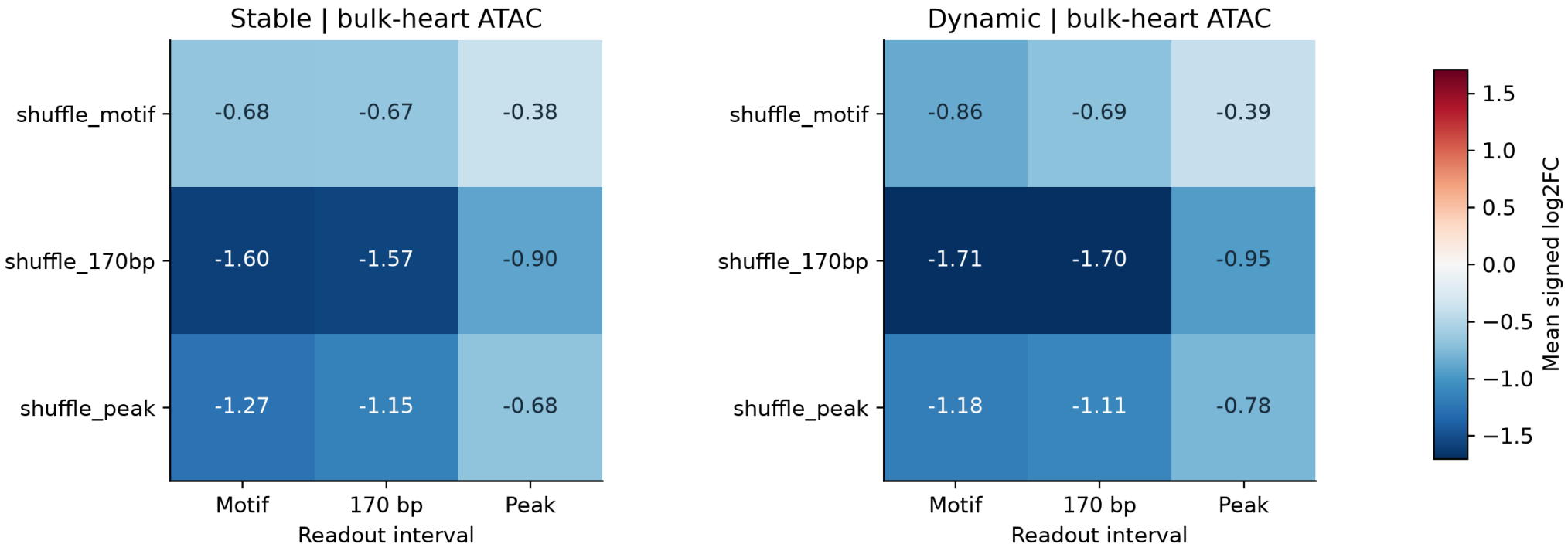


Chromatin and transcriptional responses are significantly coupled

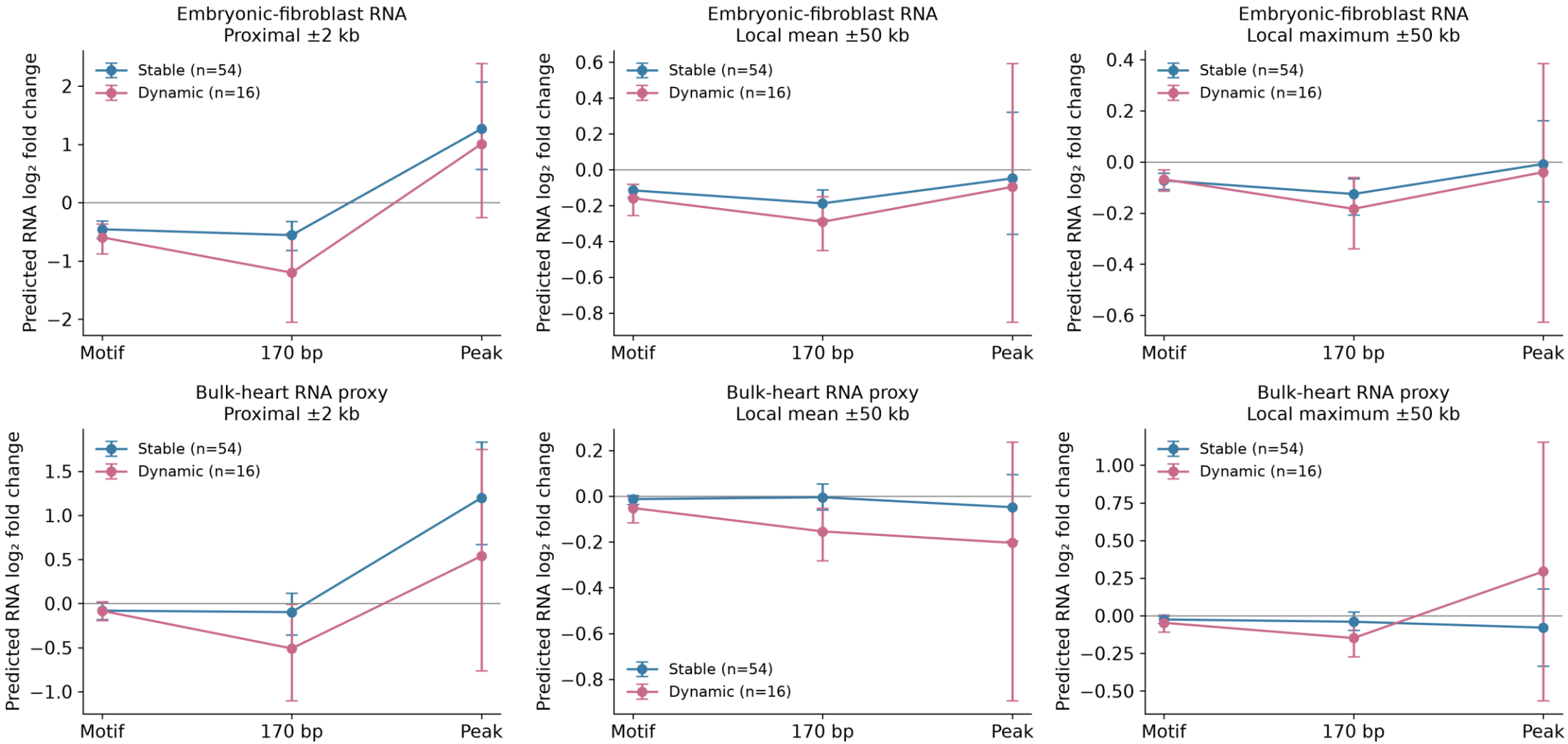


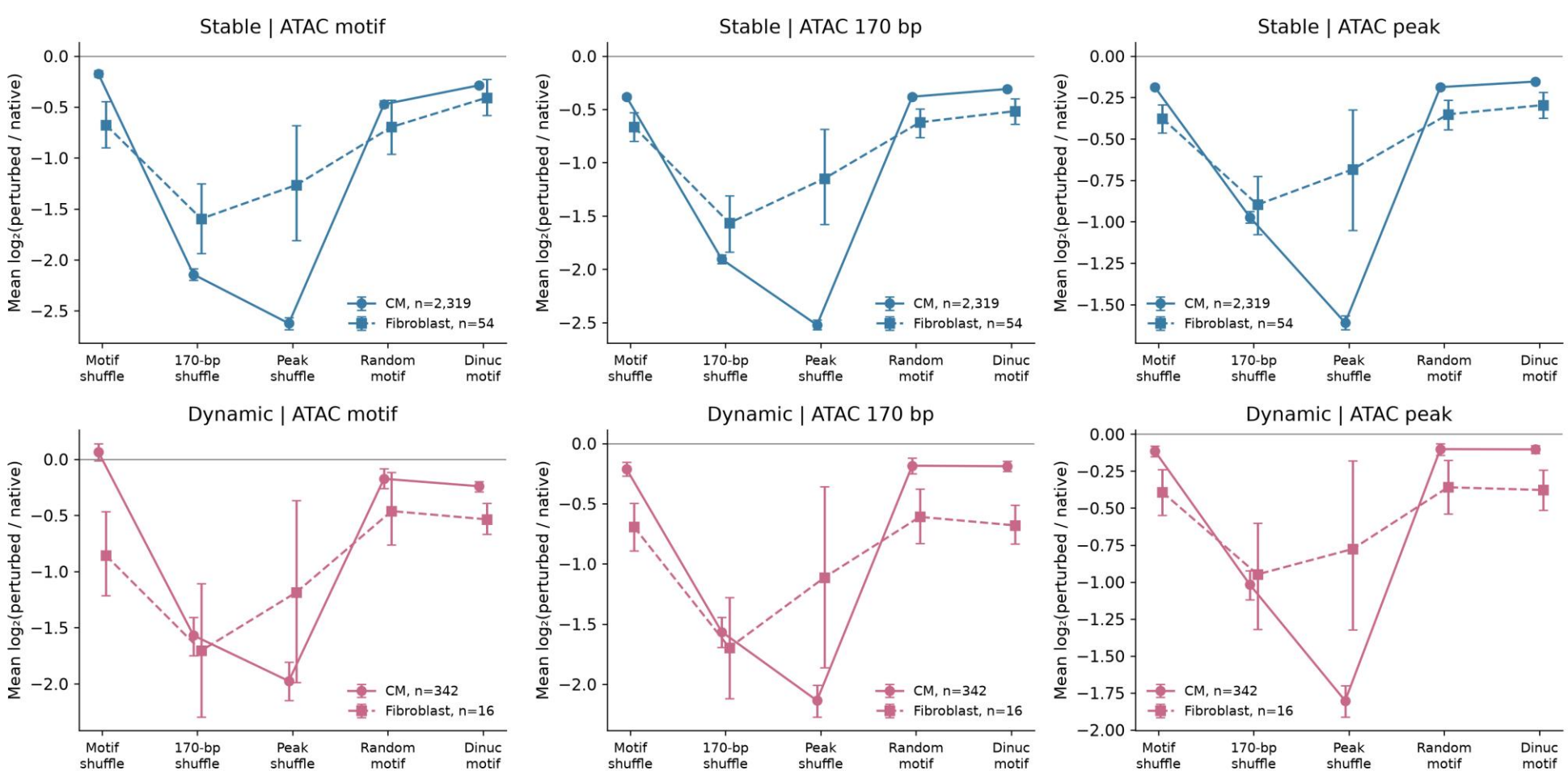


Fibroblast effects depend on edit and readout width

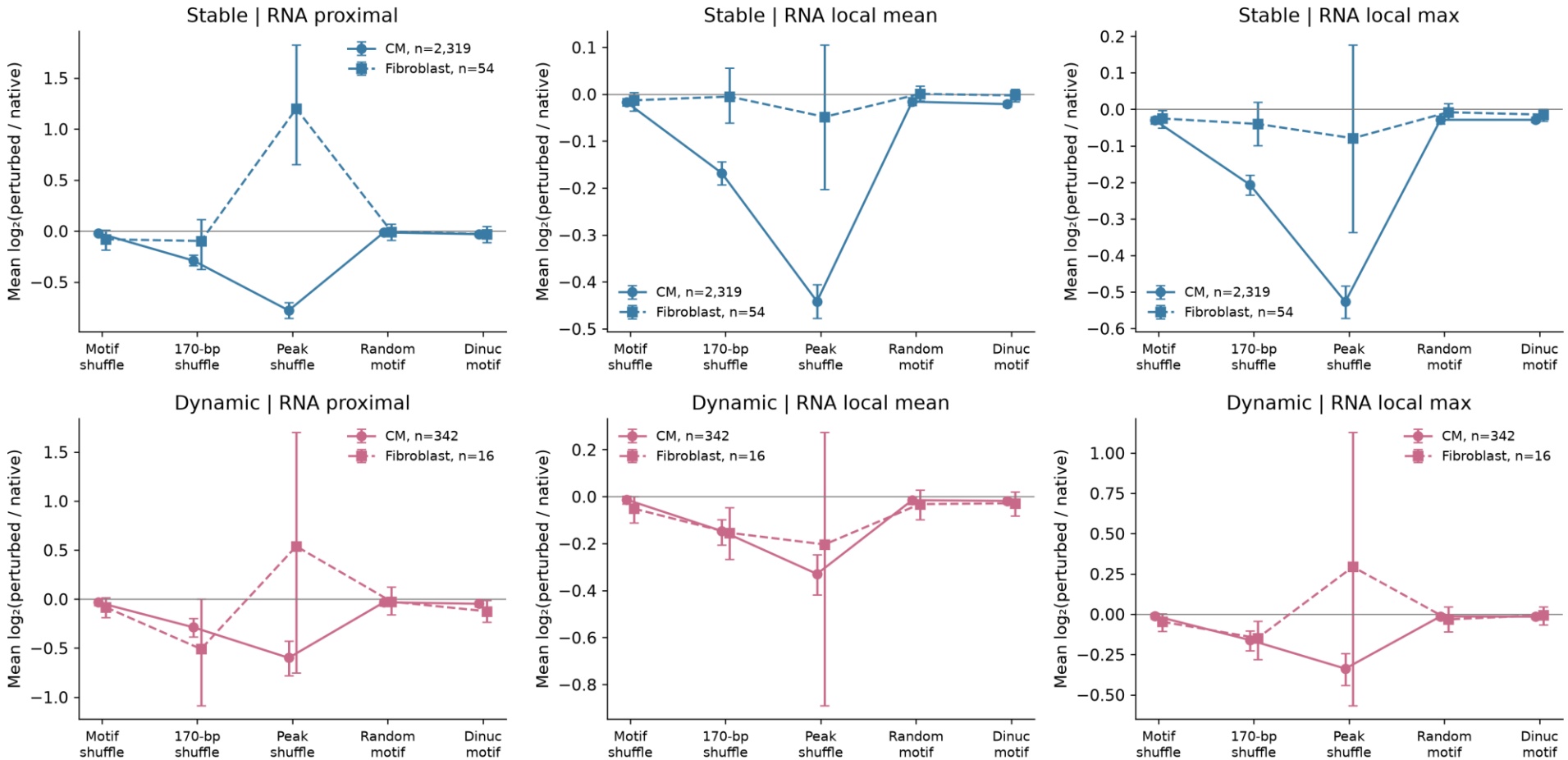


Fibroblast shuffling: RNA responses depend on edit width and prediction track



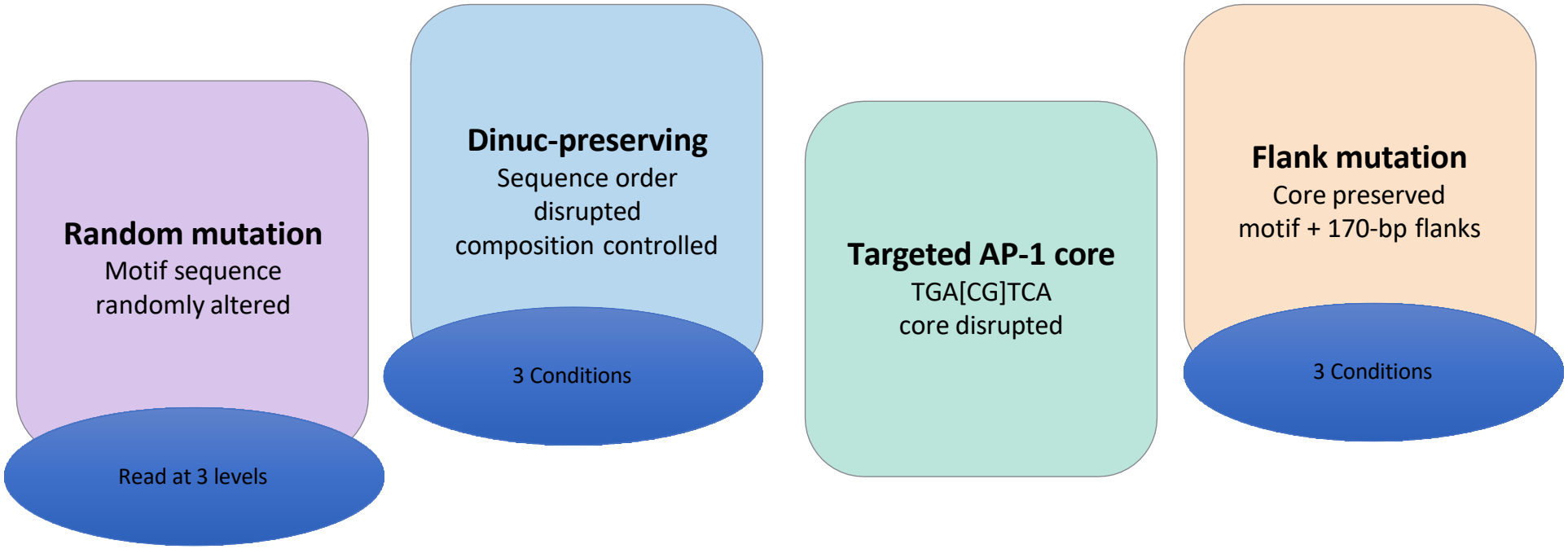

ATAC perturbation profiles across CM and fibroblast datasets
Stable | ATAC motif
Stable | ATAC 170 bp
Stable | ATAC peak
Dynamic | ATAC motif
Dynamic | ATAC 170 bp
Dynamic | ATAC peak
Mean log₂(perturbed / native)
CM, n=2,319
Fibroblast, n=54
CM, n=342
Fibroblast, n=16
Motif shuffle
170-bp shuffle
Peak shuffle
Random motif
Dinuc motif

RNA perturbation profiles across CM and fibroblast datasets

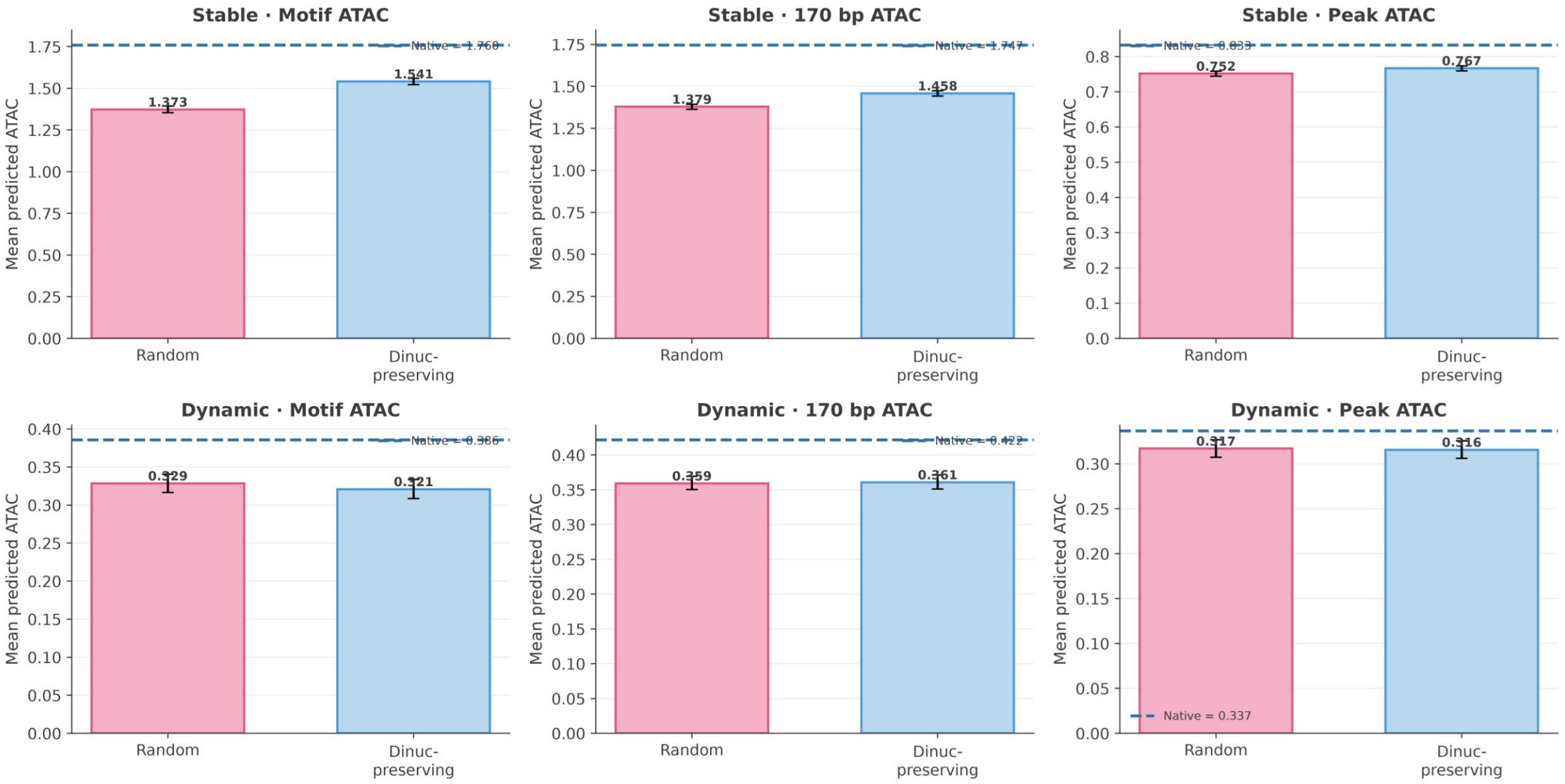


MUTATION PERTURBATIONS

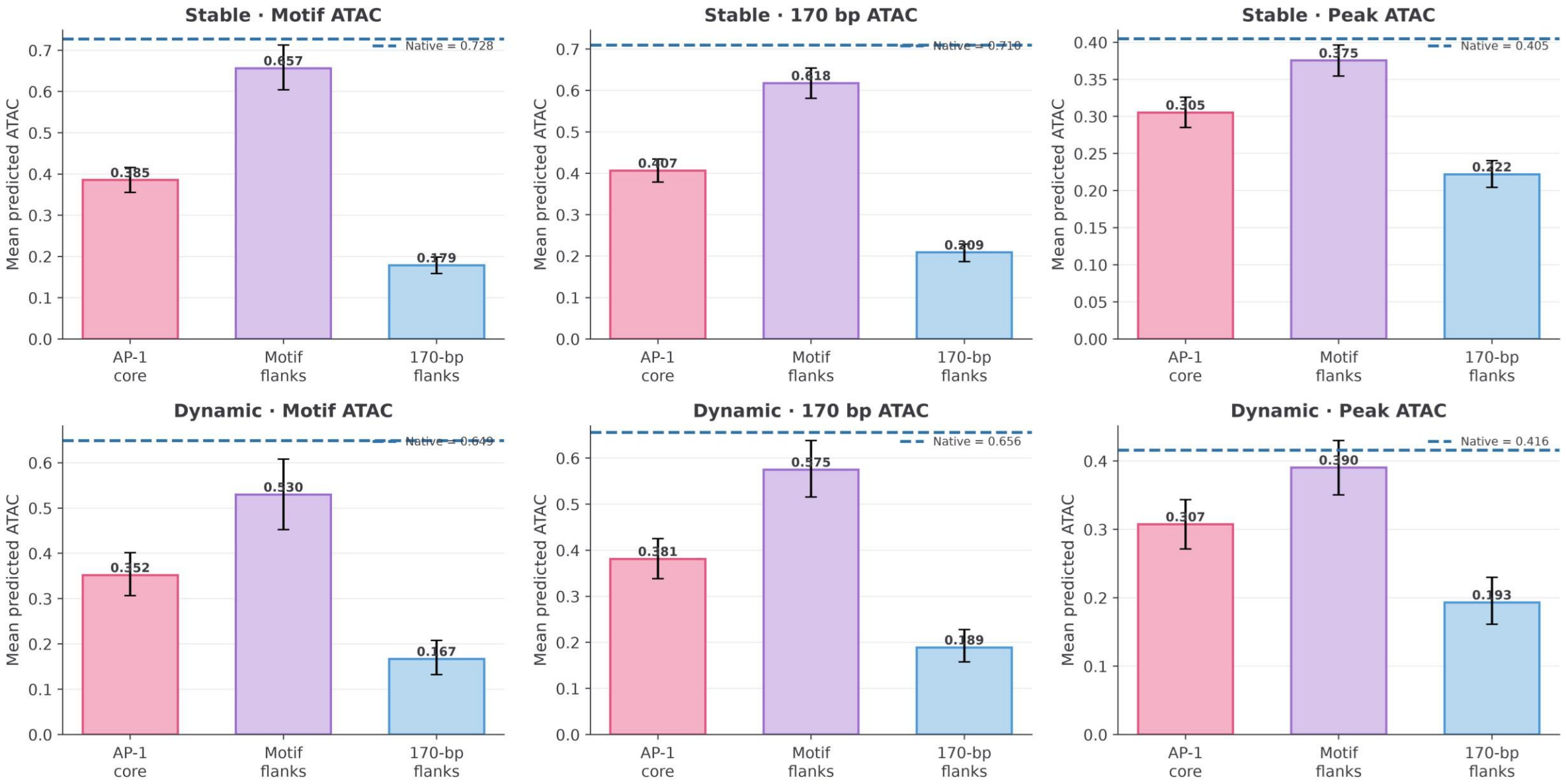
Types of Mutations



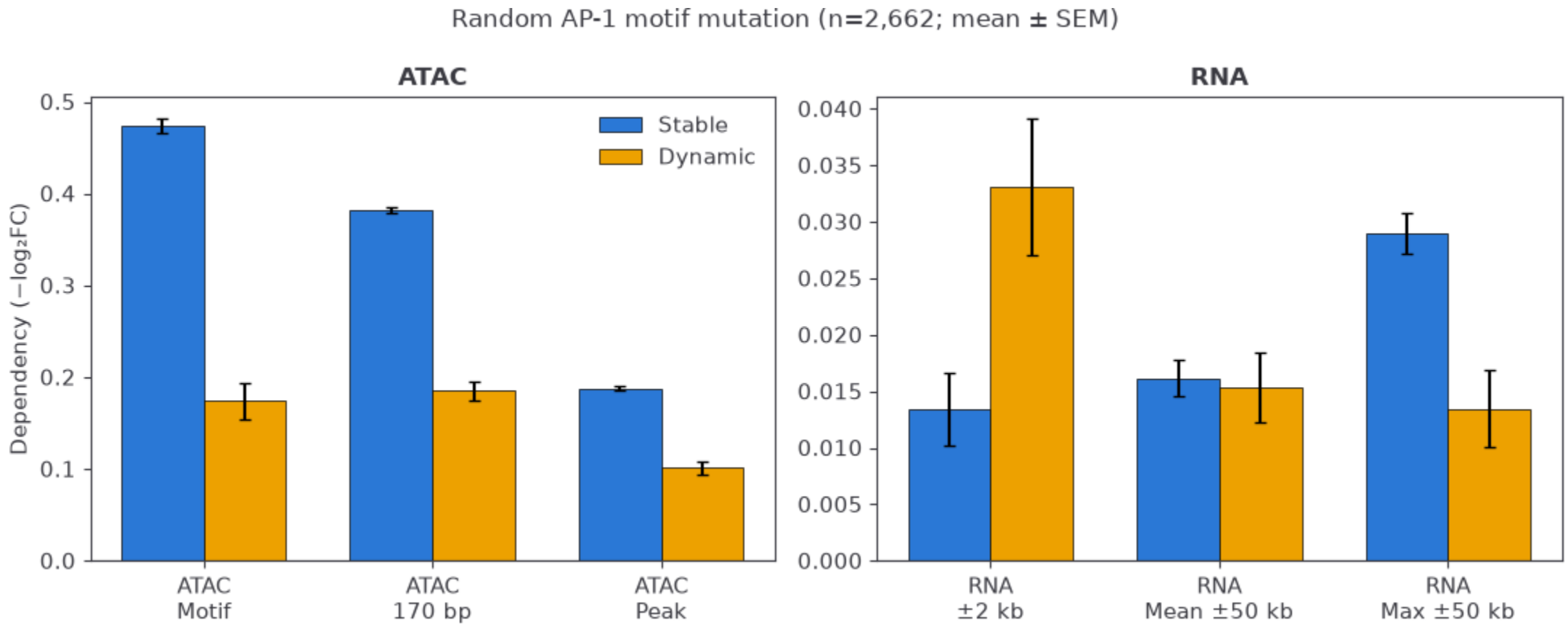
QC: mutation decreases accessibility relative to native sequence



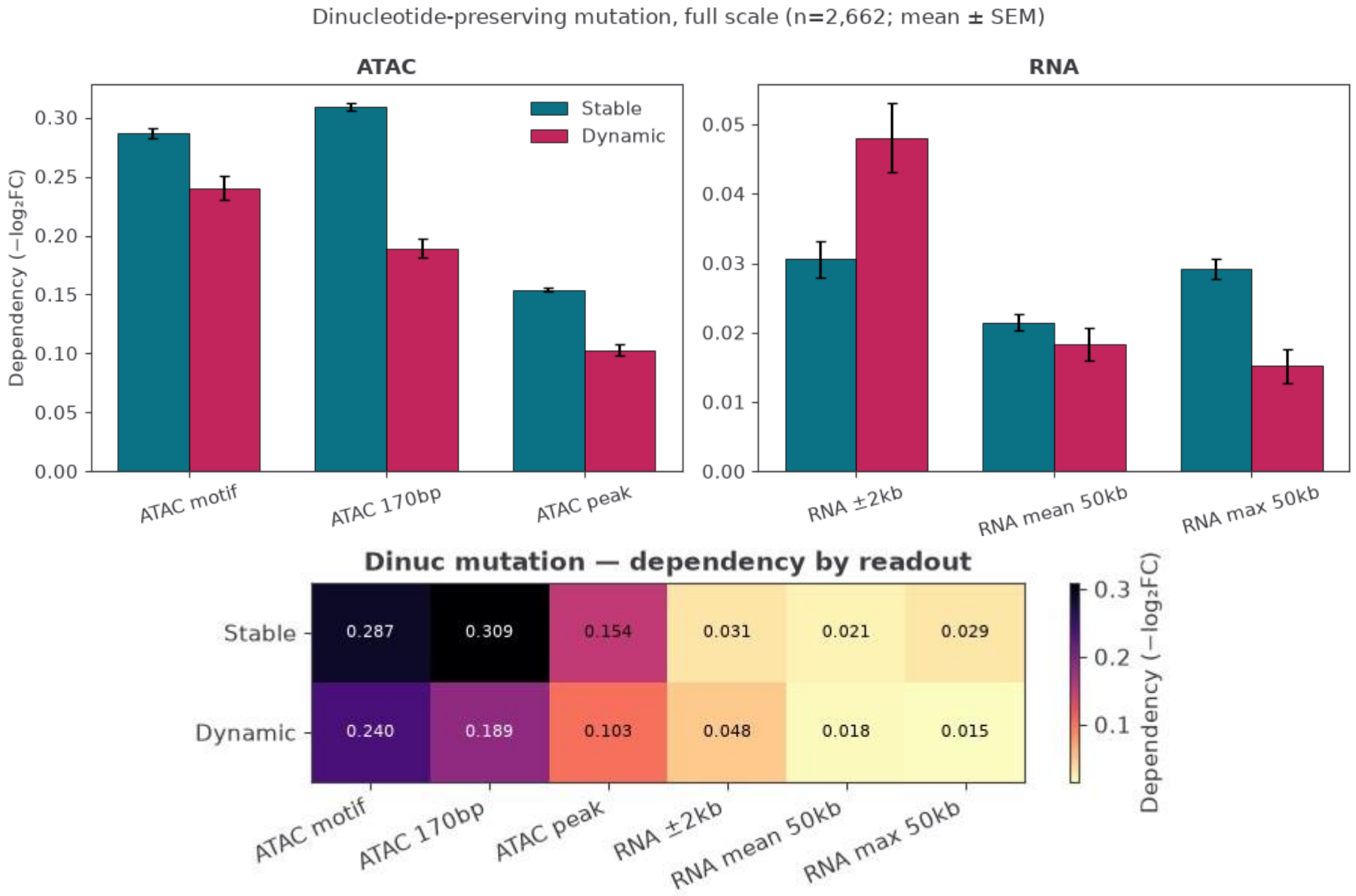
QC: targeted core and flank perturbations alter predicted accessibility



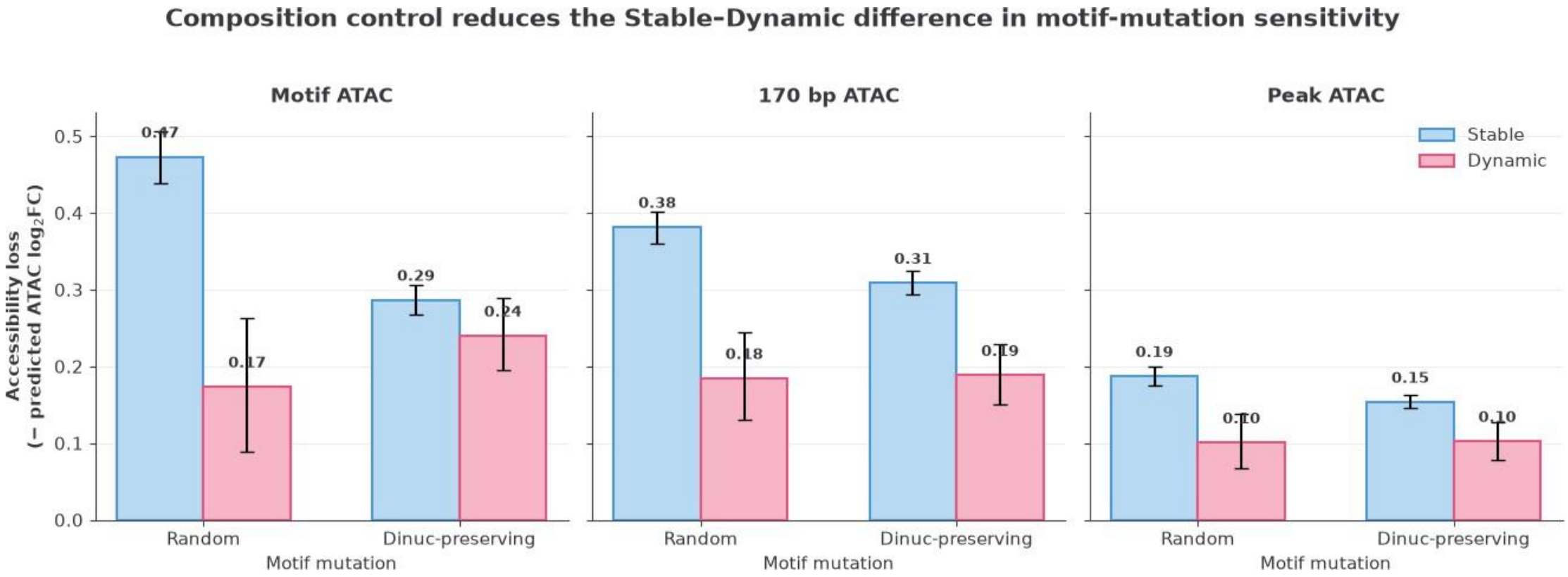
Random Mutation: ATAC and RNA reads



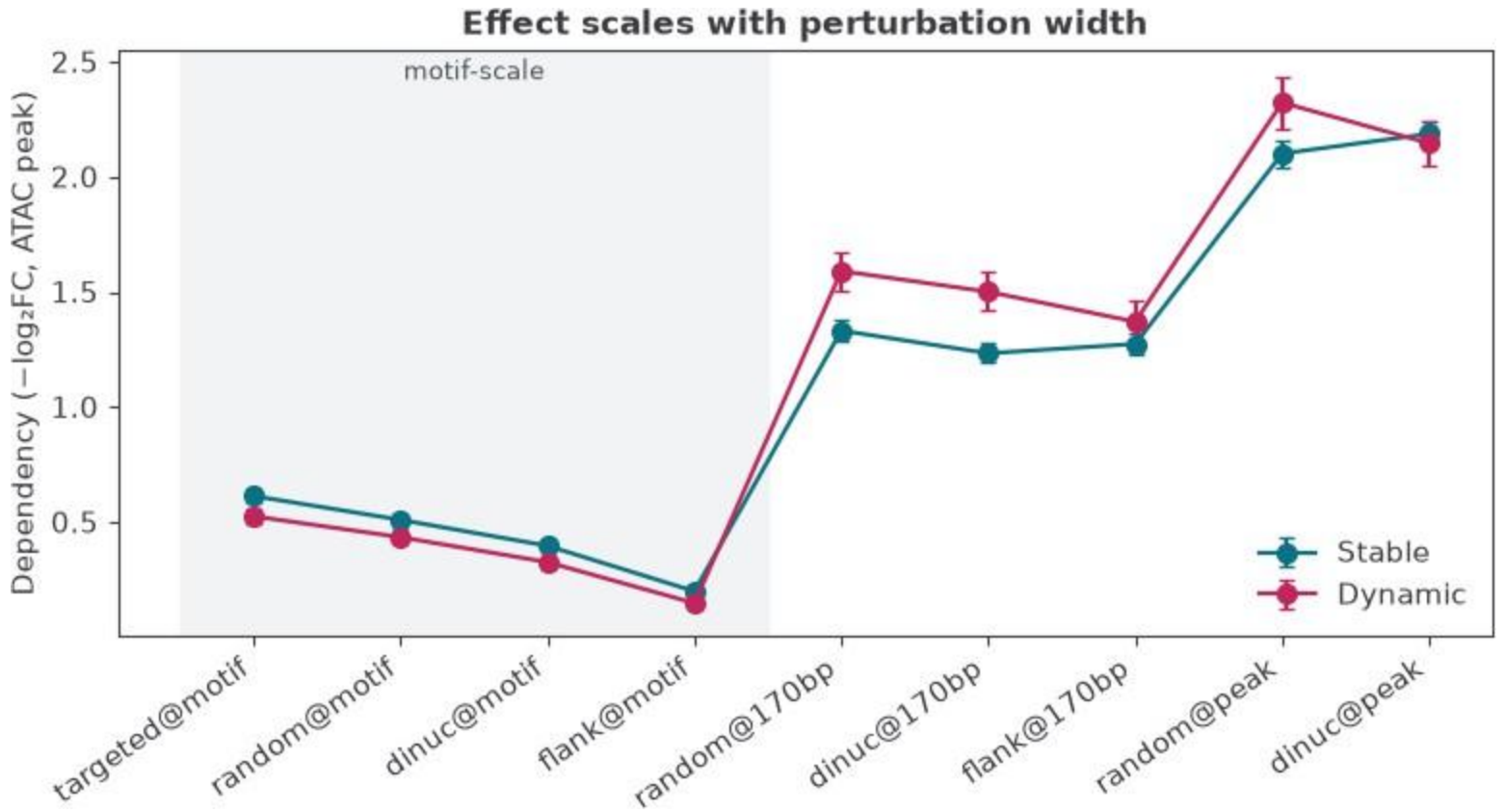
Dinucleotide Mutation: ATAC and RNA reads

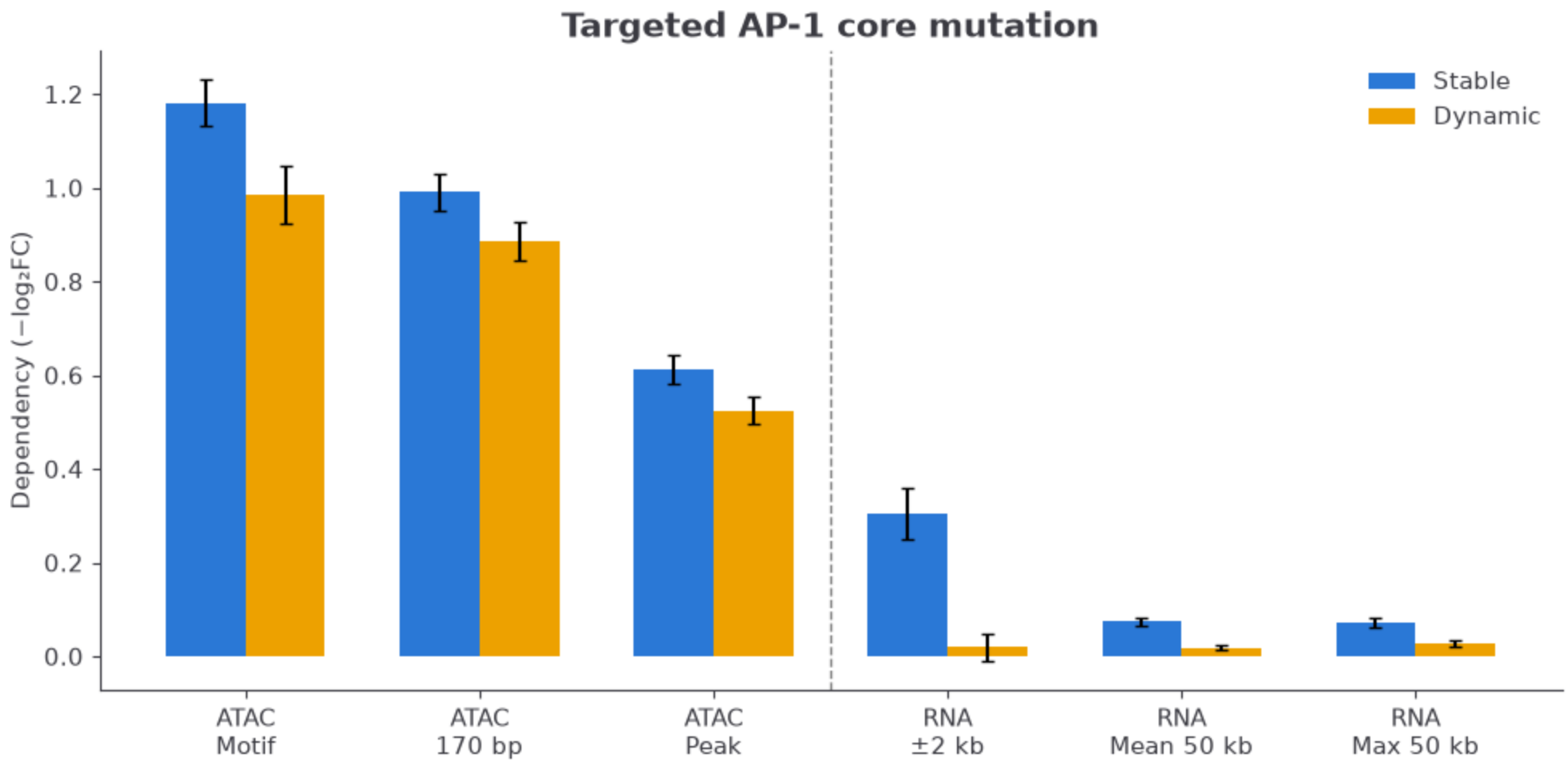


Role of Composition in accessibility

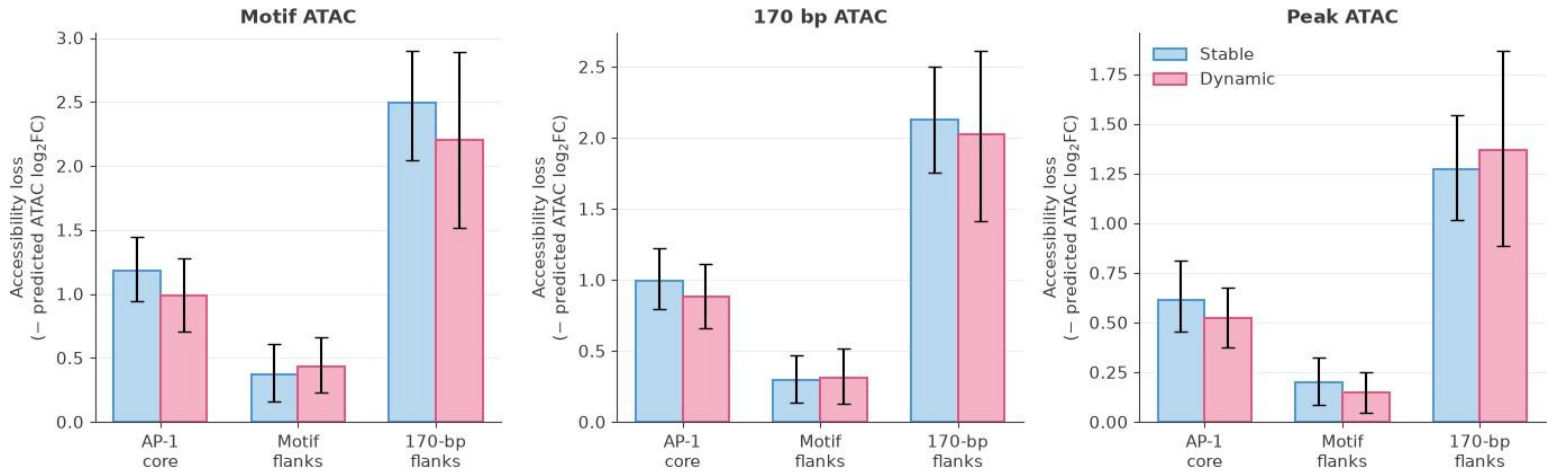


Mutation width affects peak reads

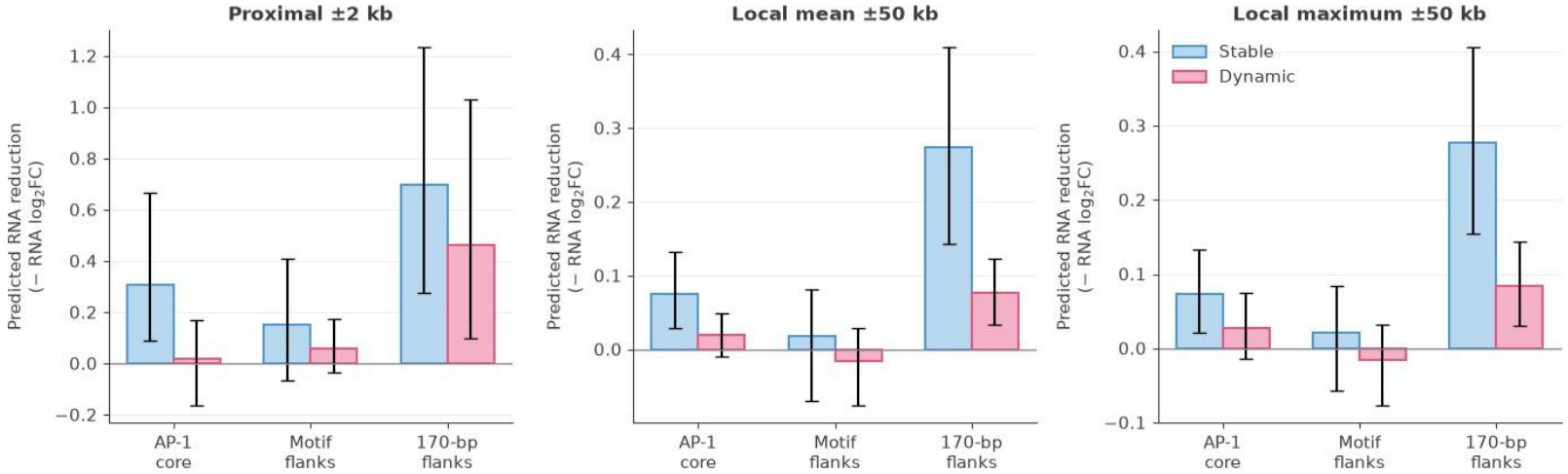



Targeted AP-1 core mutation
Stable
Dynamic
Dependency (−log₂FC)
1.2
1.0
0.8
0.6
0.4
0.2
0.0
ATAC Motif
ATAC 170 bp
ATAC Peak
RNA ±2 kb
RNA Mean 50 kb
RNA Max 50 kb

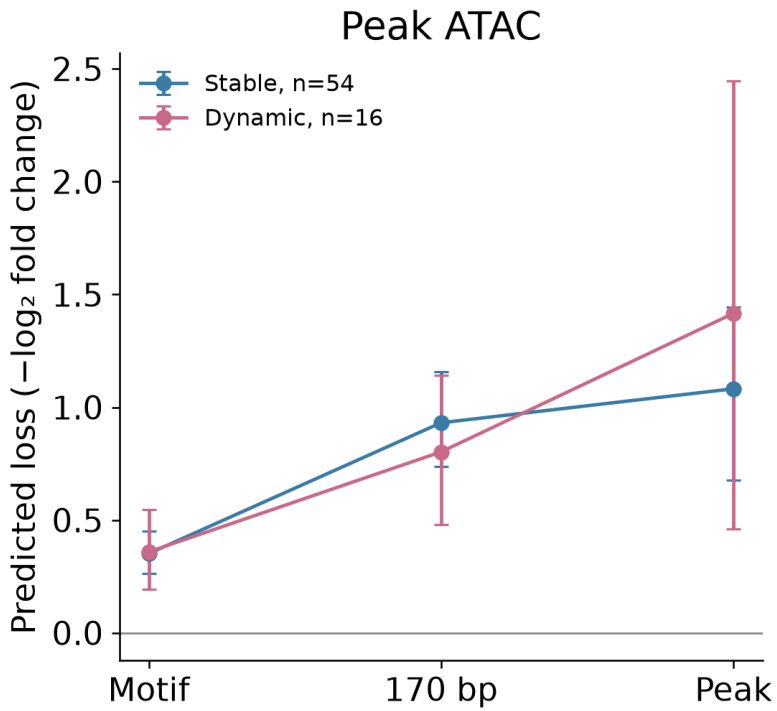
Flank Mutation

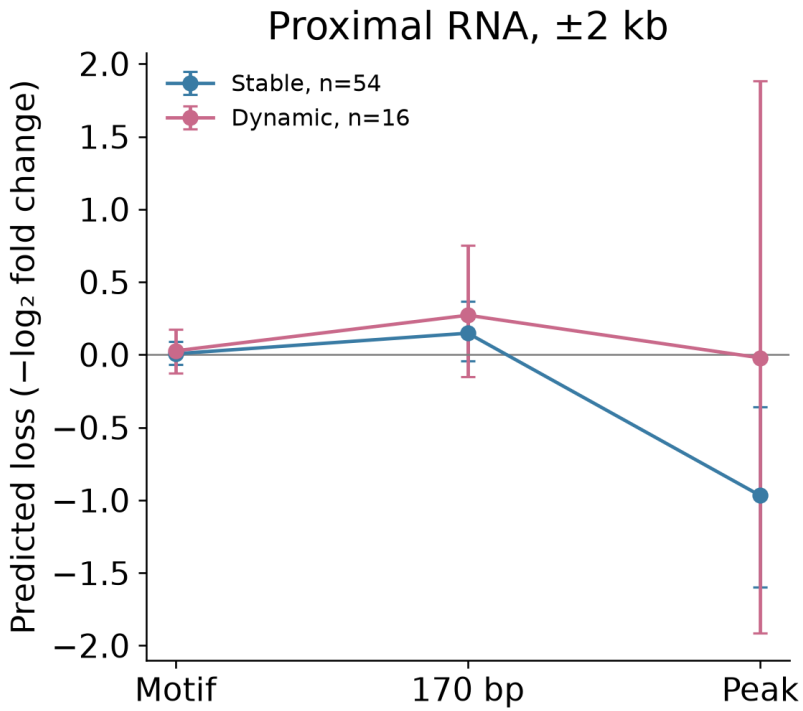


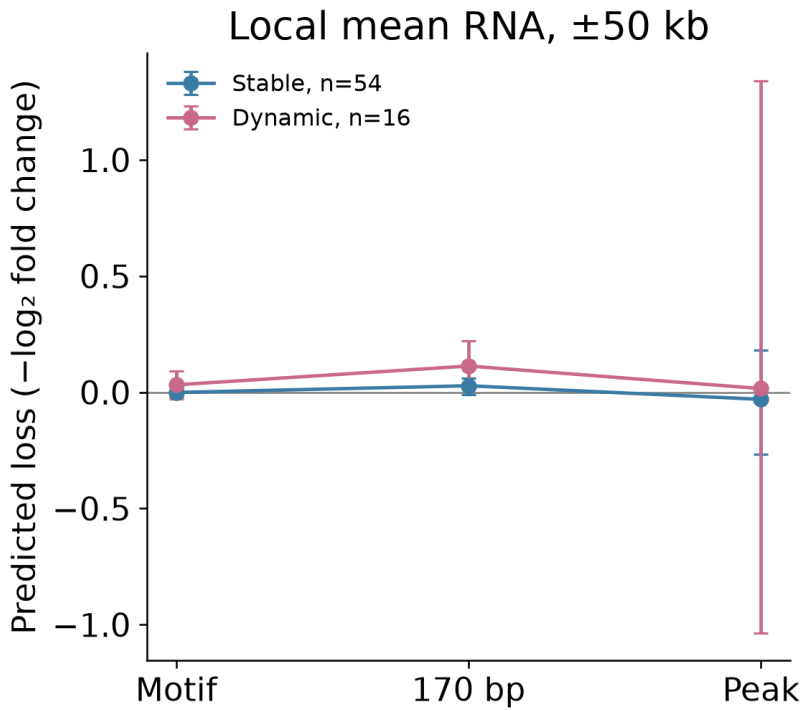
Core and flank perturbations produce distinct predicted transcriptional consequences



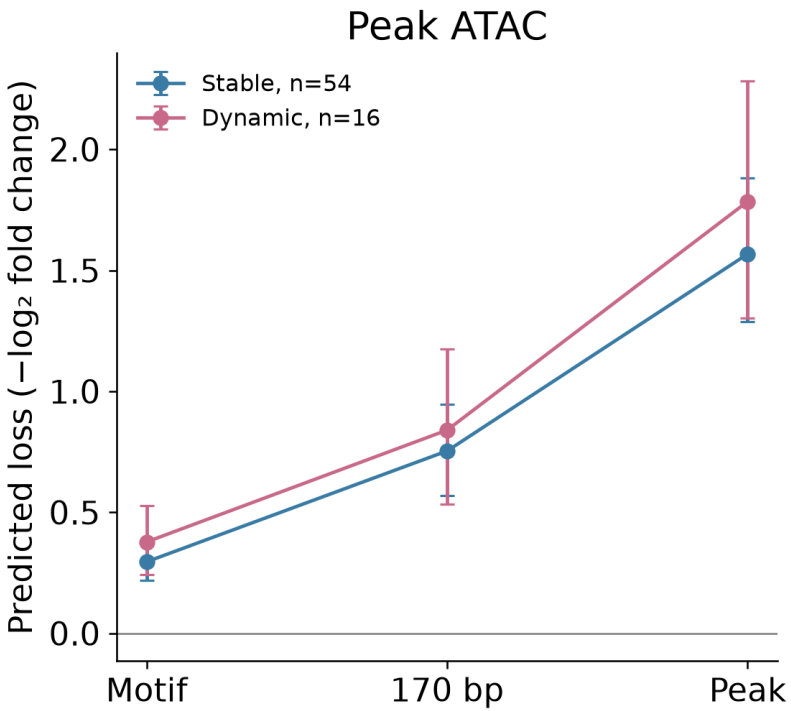
Broader random mutations

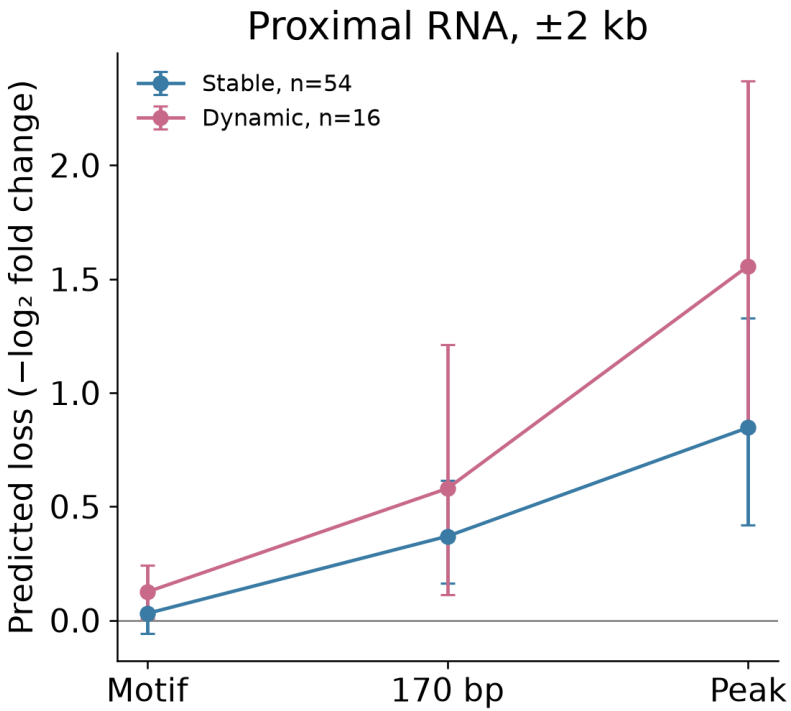


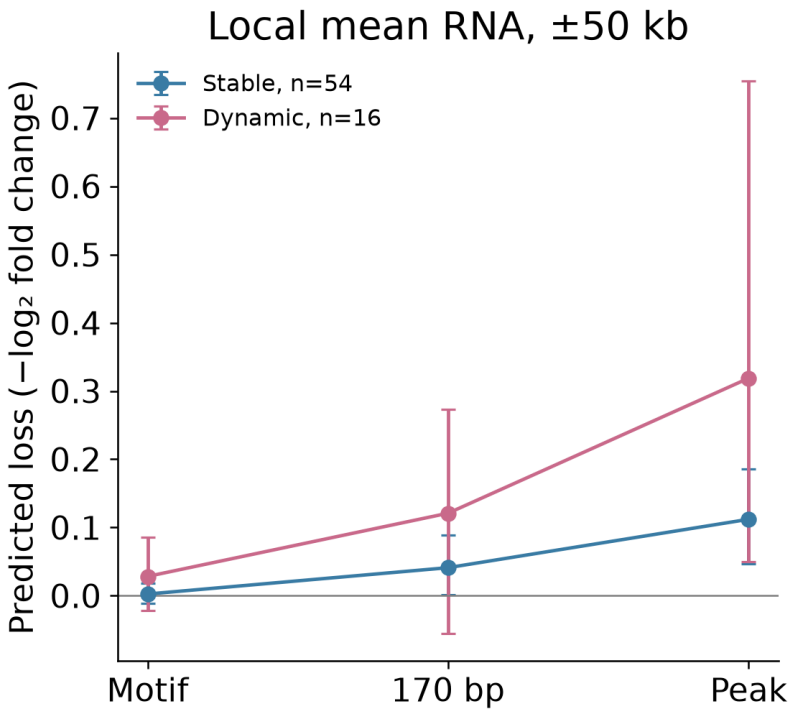




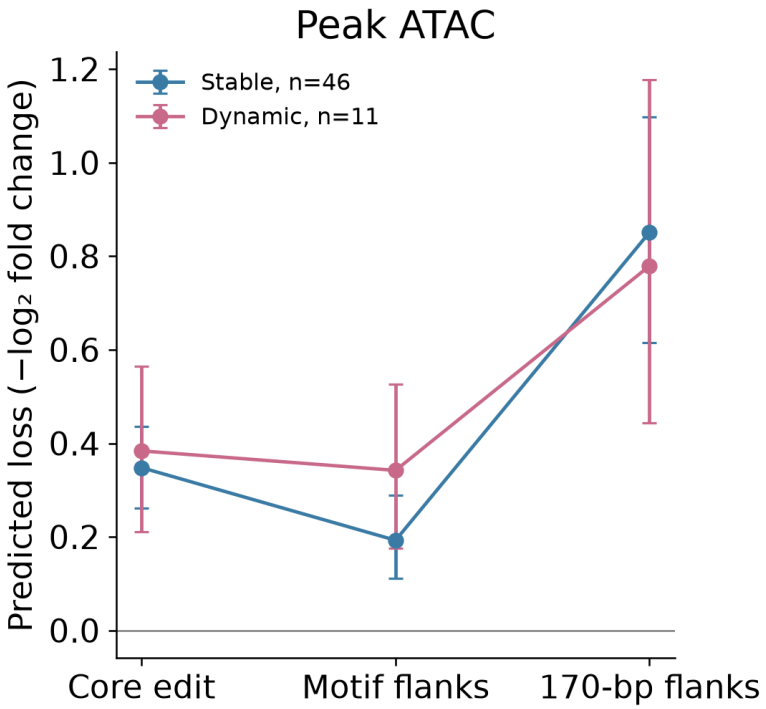
dinuc effects

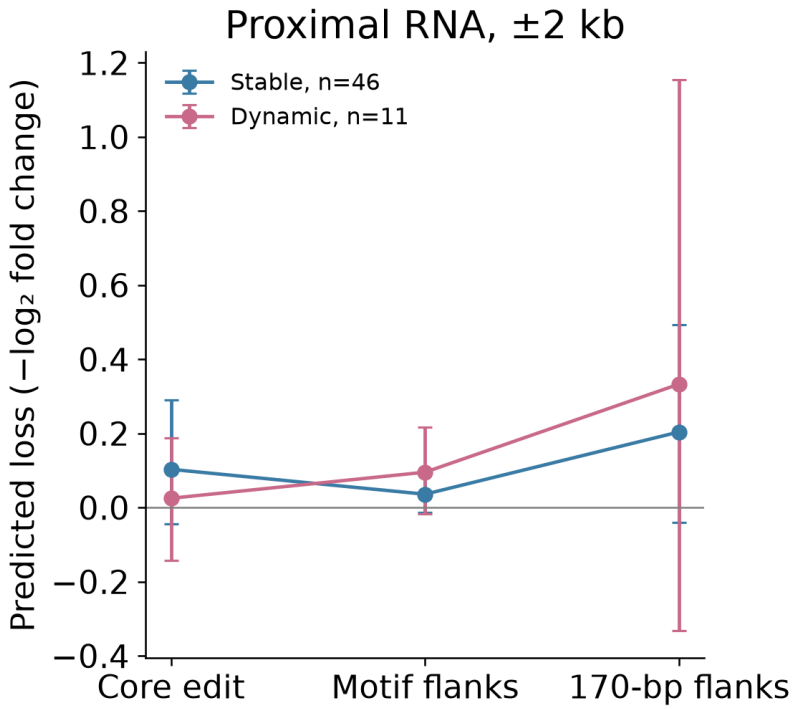


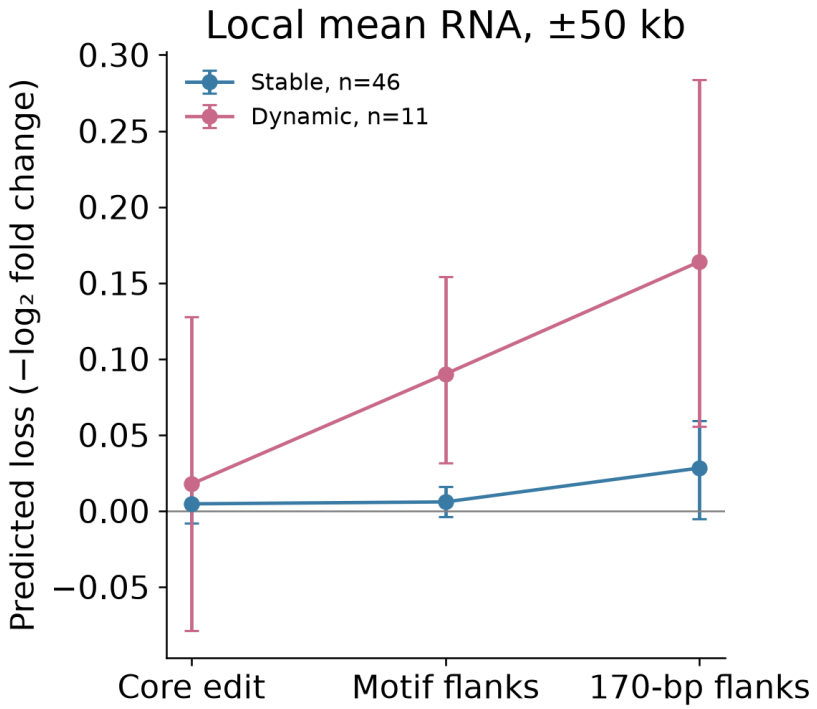




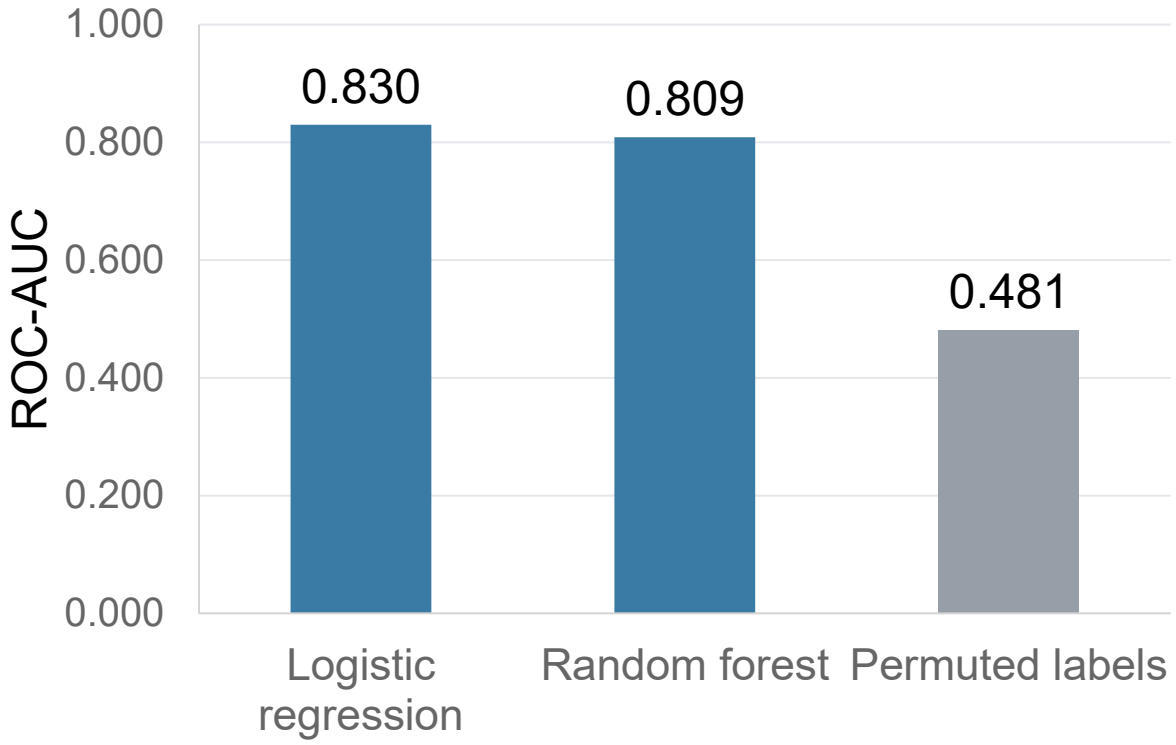
Fibroblast core and flank edits

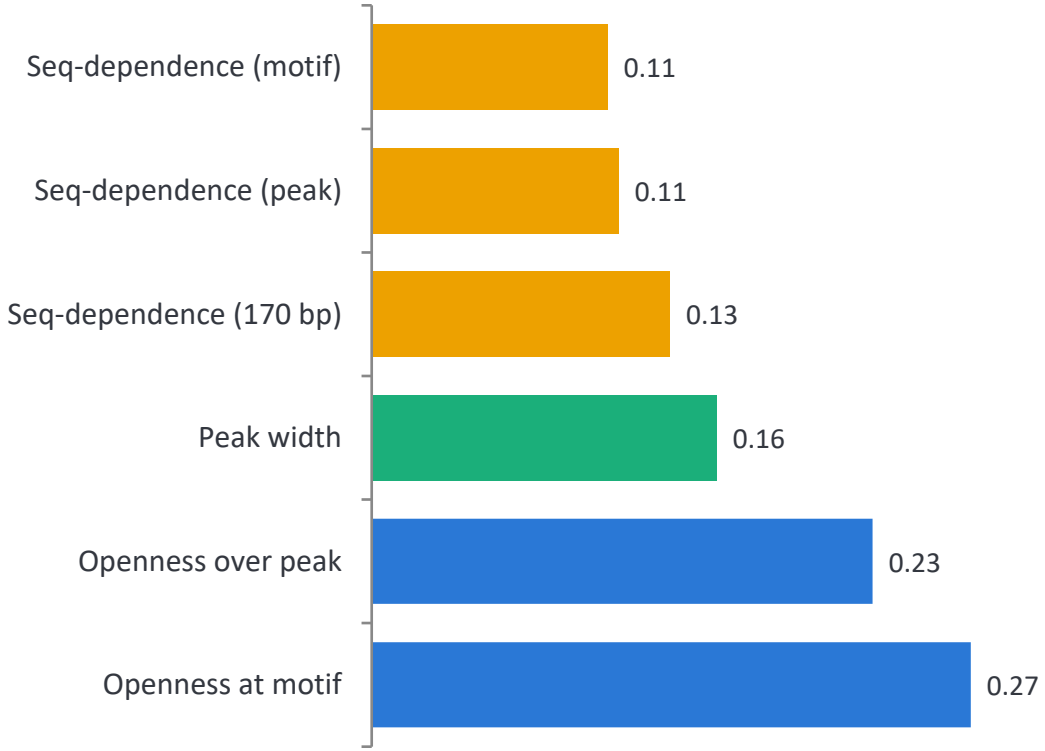






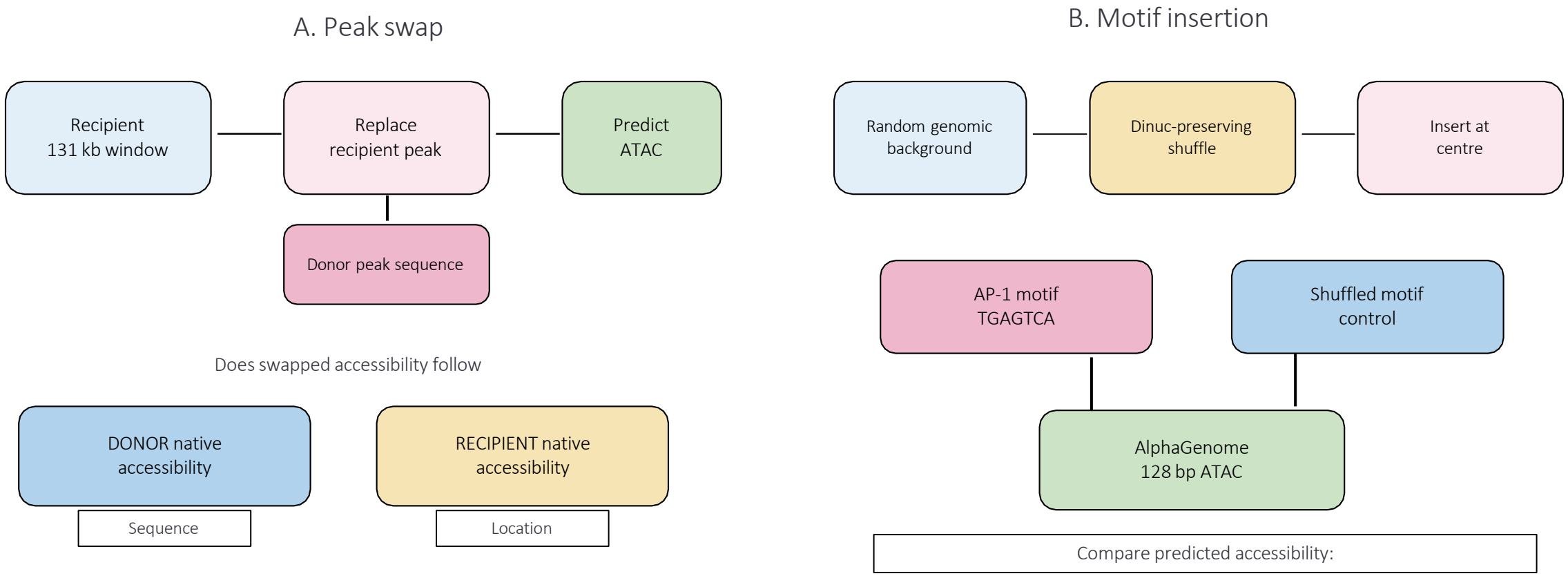
Model features distinguish Stable and Dynamic CM sites



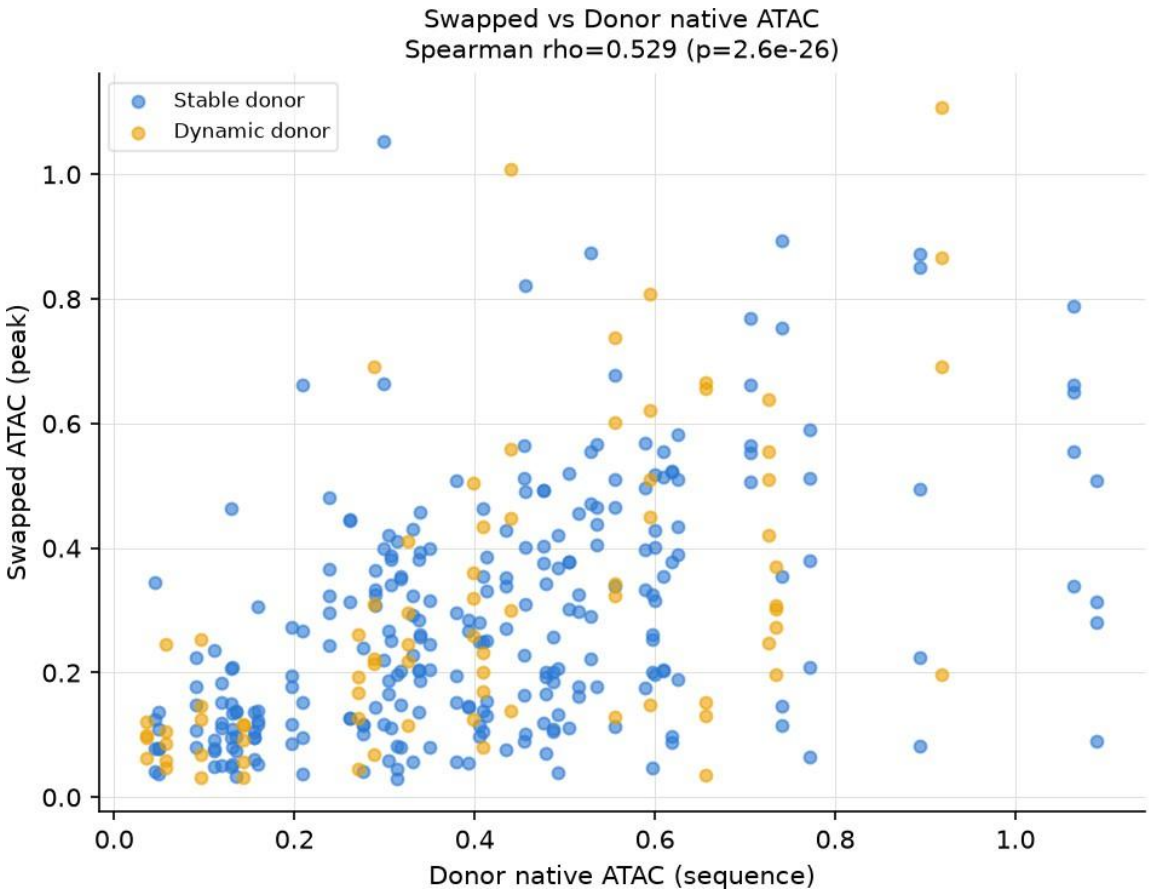


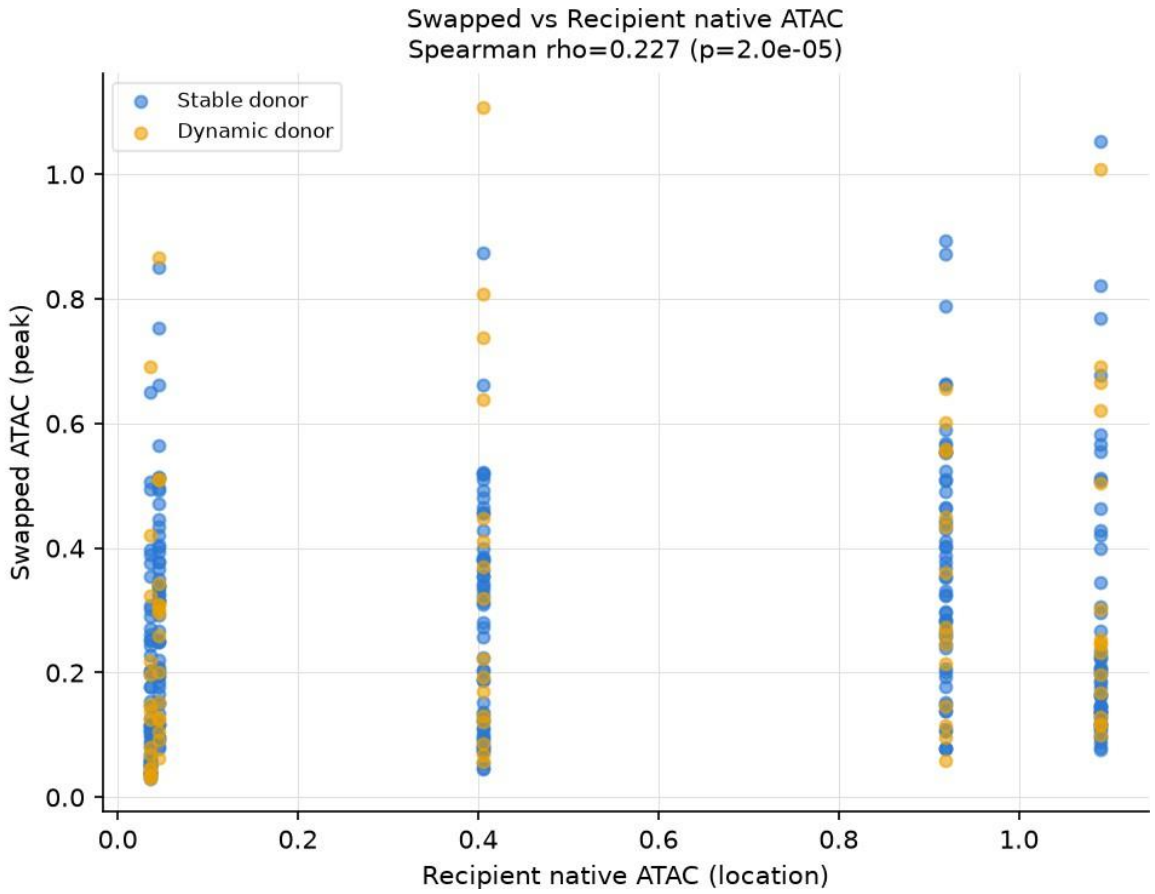
Insertion experiments

Methodology



Insertions: Peak swap





Insertion: Motif Insertion

